Deciding Without Generating: A Survey of Typed Decision Models, from Jev to Open Decision LLMs

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Abstract

Many applications of large language models need a decision rather than free text. Examples include assigning a category, giving a score, answering yes or no, or choosing the next action. Prior work has used language models for such decisions in several ways, including zero-shot classification, reward models and verifiers, LLM-as-a-Judge, safety classifiers, and model cascades. In most of these approaches, the model generates text, and the decision must then be extracted from it. Recently, a class of models has emerged that answers typed questions directly, without generating text. Given an input and a set of questions with predefined answer types, such a model returns, for each question, an answer in the declared format together with probabilities over the possible answers, and for some question types a confidence score. We call these models *typed decision models*. The commercial model Jev made this design widely known, and it was soon followed by many open models and by similar services from other providers. Research on typed decision models is growing quickly but is scattered across many communities. This survey reviews the area. We trace its roots in earlier work and give a formal description of typed decision models whose outputs depend on the question type. We analyze how these models are combined with stronger fallback models, and show that such a combination improves accuracy only if the fallback corrects more errors than it introduces on the inputs passed to it. We then review model designs, training methods, sources of efficiency, applications, benchmarks, and known failure modes. In the reported settings, typed decision models are often much faster and cheaper than generative models, and they are often competitive on tasks where the needed evidence is stated in the input. They tend to perform worse on tasks that require multi-step reasoning and against supervised models trained on in-domain data. Their probabilities also depend on how a question is asked, and they can be misled by option names and by untrusted input. We conclude with open problems for future research.

1 Introduction

Many applications of large language models (LLMs) need a decision rather than a piece of text. A support system must decide which team should handle a ticket. A content filter must decide whether a message is harmful. An agent must decide which tool to call next. An evaluator must decide which of two answers is better. The usual way to obtain such decisions is indirect. A generative model first produces text, often a chain of reasoning followed by an answer. The answer is then extracted from the text and checked against the expected format. If the model's confidence is needed, it is usually asked to state it in words, and such stated confidence is often poorly calibrated (Guo et al., 2017; Kadavath et al., 2022; Tian et al., 2023). Each of these steps adds time and cost, and each can fail.

Research has long studied how to use language models for decisions, along several separate lines. Zero-shot classifiers treated each label as a hypothesis to be tested against the input (Yin et al., 2019). In-context learners were used as classifiers by reading the probabilities of label words or option letters (Brown et al., 2020; Schick & Schütze, 2021a; Robinson et al., 2023), and later work corrected their biases toward certain labels and positions (Zhao et al., 2021; Holtzman et al., 2021; Zhou et al., 2024; Zheng et al., 2024a). Reward

models and verifiers turned preferences and correctness into scores (Christiano et al., 2017; Ouyang et al., 2022; Cobbe et al., 2021; Lightman et al., 2023). LLM-as-a-Judge used prompted models as evaluators (Zheng et al., 2023; Liu et al., 2023), and safety classifiers used them to detect harmful content (Inan et al., 2023). Model cascades and routers decided which model should handle an input (Varshney & Baral, 2022; Chen et al., 2023; Aggarwal et al., 2024; Ong et al., 2024). Selective prediction and learning to defer decided when a model should not answer at all (Chow, 1970; Geifman & El-Yaniv, 2017; Madras et al., 2018). At the same time, work on efficient inference made a single forward pass of a model much cheaper (Kwon et al., 2023; Zheng et al., 2024b; Reddy et al., 2024; Schuster et al., 2022). Each of these lines addressed part of the problem. None of them, however, offered a general model that answers arbitrary typed questions directly with probabilities over the possible answers.

Recently, a class of models has emerged that addresses this gap. We call them *typed decision models*. Such a model receives an input, which we call the *state*, and a set of questions. Each question has a predefined answer type. It may ask the model to choose one of several options (CHOICE), to rate the input on an ordered scale (SCORE), or to give the probability that a statement is true (NOUL). The output depends on the type (Figure 1). For a CHOICE question, the model returns the selected option, a probability for each option, and a confidence score. For a SCORE question, it returns a probability for each level, the expected level, and a confidence score. For a NOUL question, it returns only the probability that the statement is true. The model does not generate text, and because the possible answers are fixed in advance, its outputs are always in the declared format. Whether its probabilities are also calibrated is an empirical question, which we examine in Section 9.4.

The commercial model Jev, released by TypeSafe AI, made this design widely known (Almeida, 2026; TypeSafe AI, 2026b). Its developers call it a "System One" model. The name refers to the distinction in psychology between fast, intuitive thinking (System 1) and slow, deliberate thinking (System 2) (Kahneman, 2011). The term describes the intended use: quick decisions inside software, with harder cases passed to slower reasoning models. It does not refer to a specific technique, and in this survey we use the neutral term *typed decision model* instead. According to its developers, Jev produces all answers in a single inference call without generating tokens one by one, and is trained with a method called *Reinforcement Learning for Calibrated Decisions* (RLCD) (Almeida, 2026). Its weights and training details have not been released (Li & Wagle, 2026).

Jev attracted wide attention. More than one hundred studies have analyzed, extended, or applied it (Tang & Zheng, 2026; Ling et al., 2026). Researchers have released dozens of open models with the same design (Li & Wagle, 2026; ConvAI Innovations, 2026; apolinario, 2026). Other providers have also adopted it: Cloudflare released open models that answer the same kinds of questions (Cloudflare, 2026), and OpenAI introduced a service with three matching question types (OpenAI, 2026). Typed decision models have been used as evaluators (Li et al., 2026i; Rao & Callison-Burch, 2026), as controllers in agents (Wu & Lim, 2026b; Wang, 2026), for network management (Li et al., 2026c;d), for content moderation (Mazzoni & Failla, 2026; Paschalides et al., 2026), for security (Wang & Gao, 2026; Severo et al., 2026), and for labeling large text collections (Rafe & Das, 2026a; Denney & DiGiuseppe, 2026; Ibrahim & Zaki, 2026). Other studies have attacked these models (Wu & Lim, 2026a; Xu et al., 2026b; Hu et al., 2026), examined their probabilities (Li et al., 2026e; Joy, 2026; Shankaranarayana et al., 2026), and shown that ordinary LLMs can be used in the same way without changing their architecture (Li & Wagle, 2026; Zhang et al., 2026b).

This growing body of work is scattered across machine learning, natural language processing, networking, security, and the social sciences. It uses inconsistent terms: the same kind of model is called a typed decision model (Tang & Zheng, 2026), a semantic decision engine (Li et al., 2026b), a System One model, or a Jev-style model. The studies also differ widely in their methods and findings. A survey is therefore needed to organize this area. In addition, most ideas behind typed decision models have a long history. Relating the new work to this history helps to separate what is new from what is not, and allows the new area to build on established methods. Finally, several providers and many open models now share the same design. We therefore focus on typed decision models as a class, defined by their interface rather than by a particular architecture. Jev is the model studied in most of the surveyed work, partly because many studies were motivated by its release. Throughout, we use *Jev* for the commercial model and its versions, *Jev-style models* for open models that

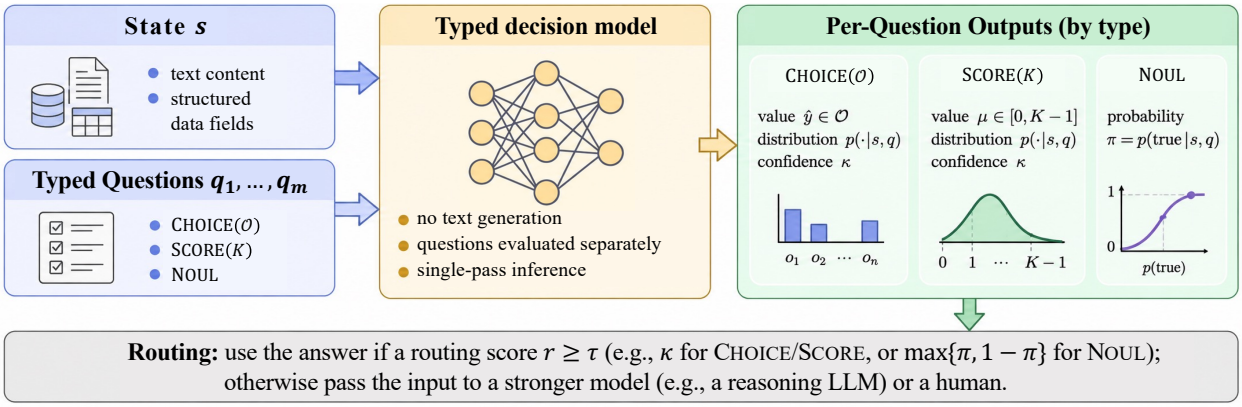


Figure 1: Overview of a typed decision model. The model receives an input (the state) and a set of typed questions. Its output depends on the question type: a selected option ($\hat{y}$), the expected level (μ), or the probability of "true" (π), together with answer probabilities (p) and, for CHOICE and SCORE, a reported confidence (κ). A routing score decides whether the answer is used directly or the input is passed to a stronger model. Most applications in Section 7 follow this pattern.

follow its question types, and *decision LLMs* for general-purpose LLMs that are used as typed decision models, for example by reading the probabilities of their next token.

Contributions.

1. We review the earlier research that typed decision models build on (Section 2).
2. We give a formal description of typed decision models whose outputs depend on the question type (Section 3). We distinguish well-formed, correct, and calibrated answers, and we separate answer probabilities, reported confidence, and the scores used for routing. We also characterize when combining a typed decision model with a stronger fallback improves accuracy (Proposition 1).
3. We review model designs (Section 4), training methods (Section 5), sources of efficiency (Section 6), applications (Section 7), evaluation (Section 8), and failure modes (Section 9).
4. We discuss what the current evidence supports (Section 10) and outline open problems (Section 11).

2 Foundations

Typed decision models combine ideas from several earlier lines of research. Each of these lines already used language models to make decisions. This section reviews them in turn. Methods for efficient inference are reviewed separately in Section 6. Table 1 summarizes how each line relates to typed decision models.

2.1 Zero-shot classification and label probabilities

Before instruction-tuned LLMs, zero-shot text classification was often cast as textual entailment: each label was turned into a hypothesis and tested against the input (Yin et al., 2019). Later work trained general zero-shot classifiers on entailment data (Laurer et al., 2023) and built small bidirectional encoders that score labels or entity types given at inference time (Zaratiana et al., 2024; Stepanov et al., 2025). These encoders are direct predecessors of the encoder-based typed decision models in Section 4.3. With GPT-3, classification could be done by reading the probability that the model assigns to each label word (Brown et al., 2020). Pattern-exploiting training made the mapping from labels to words explicit (Schick & Schütze, 2021a), and showed together with related work that small models can be competitive few-shot classifiers (Schick & Schütze, 2021b; Gao et al., 2021). These label probabilities were soon found to be biased toward certain

labels, which led to methods for calibrating them (Zhao et al., 2021; Zhou et al., 2024). Part of the problem is that different wordings of the same answer compete for probability (Holtzman et al., 2021). Scoring the input given each label, instead of each label given the input, reduces this problem (Min et al., 2022). Moreover, in instruction-tuned models the most probable first token does not always match the answer that the model gives in text (Wang et al., 2024). Probabilities and generated answers are therefore not interchangeable. In multiple-choice questions, LLMs also prefer certain option letters or positions regardless of content (Zheng et al., 2024a; Pezeshkpour & Hruschka, 2024). Two lessons from this work apply directly to typed decision models. First, the name of a label carries information that can override its definition. Second, raw label probabilities need calibration. Both issues appear again in recent studies (Section 9).

2.2 LLM-as-a-Judge

Strong LLMs used as judges agree with human preferences about as often as humans agree with each other (Zheng et al., 2023). Judges guided by a rubric extend this to detailed criteria (Liu et al., 2023; Kim et al., 2024). LLM judges have known biases, for example toward the first answer shown, toward longer answers, and toward their own outputs (Zheng et al., 2023; Wang et al., 2023). They usually write an explanation before giving a verdict. Earlier surveys cover this area (Gu et al., 2024; Li et al., 2024). A typed decision model can be seen as a judge that gives a verdict without an explanation. Several recent studies ask whether this trade-off is acceptable (Li et al., 2026i; Rao & Callison-Burch, 2026; Wang & Gao, 2026).

2.3 Reward models and verifiers

Reward models turn human preferences into a score that is used to train a model (Christiano et al., 2017; Ouyang et al., 2022). They are usually trained on pairwise comparisons under the Bradley–Terry model (Bradley & Terry, 1952). Verifiers decide whether a solution, or a step in a solution, is correct (Cobbe et al., 2021; Lightman et al., 2023). Generative verifiers use the probability of the token "Yes" as a score (Zhang et al., 2024), which is essentially a NOUL question. Reward models are evaluated by how often they prefer the better answer (Lambert et al., 2024), and they can be exploited when a model is optimized against them (Gao et al., 2023). The training method of Jev, RLCD, is presented as an alternative to training from human preferences and from verifiable rewards (Almeida, 2026; DeepSeek-AI et al., 2025).

2.4 Safety classifiers

LLMs have been fine-tuned to classify prompts and responses according to a list of safety categories. The decision is read from the probability of an "unsafe" token (Inan et al., 2023). Such safety classifiers are an early form of typed decision model. Their categories, however, are fixed during training, whereas typed decision models accept new categories at inference time. Typed decision models are now also evaluated as safety filters (Sections 7.4 and 9.3).

2.5 Tool and action selection

Models that use tools must decide whether to call a tool and which one to call (Schick et al., 2023). Agents must choose among possible actions at every step. Generative agents express these choices as text, which is then parsed. Choosing from a finite list of actions is exactly a CHOICE question. This explains why agent control is one of the main applications of typed decision models (Section 7.3). Some recent models score candidate actions directly against the internal state of an LLM (Zhan et al., 2026).

2.6 Constrained generation

Constrained decoding ensures that generated text follows a grammar or a schema (Scholak et al., 2021; Geng et al., 2023; Willard & Louf, 2023), and specialized systems make it efficient (Beurer-Kellner et al., 2024; Dong et al., 2025; Zheng et al., 2024b). Constraints can, however, distort the model's distribution over outputs (Park et al., 2024), and strict output formats can harm reasoning (Tam et al., 2024; Banerjee et al., 2025). The latter finding foreshadows a limitation of typed decision models discussed in Section 9.6. It

guarantees the format of the output, but it does not reduce the cost: the model still generates one token at a time. Probabilities of complete answers can be obtained by scoring each candidate answer as a sequence and normalizing over the candidates, but this requires scoring every candidate and does not by itself yield calibrated probabilities. Typed decision models make this distribution the primary output. They restrict the output to a small set of answers and return probabilities over that set directly.

2.7 Calibration and selective prediction

Calibrating the scores of classifiers has a long history (Zadrozny & Elkan, 2002), and modern neural networks are often overconfident (Guo et al., 2017). LLMs are reasonably well calibrated on some multiple-choice formats (Kadavath et al., 2022), but the confidence they state in words is often poor after instruction tuning (Tian et al., 2023; Xiong et al., 2024). Geng et al. (2024) survey confidence estimation for LLMs. Calibration is measured with proper scoring rules (Brier, 1950; Gneiting & Raftery, 2007) and with the expected calibration error (Naeini et al., 2015). Selective prediction lets a classifier abstain on uncertain inputs, which reduces errors at the cost of answering fewer inputs (Chow, 1970; Geifman & El-Yaniv, 2017). Its benefits can shrink when the test data differ from the training data (Kamath et al., 2020). Conformal prediction gives such methods statistical guarantees (Angelopoulos & Bates, 2021). LLM judges can also pass uncertain cases to stronger judges while guaranteeing a level of agreement with humans (Jung et al., 2024). The confidence score that typed decision models return for CHOICE and SCORE questions is intended for the same purpose (TypeSafe AI, 2026b).

2.8 Cascades, routing, and learning to defer

Model cascades first query a cheap model and pass the input to a stronger model when the cheap model is uncertain (Varshney & Baral, 2022; Chen et al., 2023; Aggarwal et al., 2024). Jitkrittum et al. (2023) analyze when deferral based only on the first model's confidence is sufficient. They show that it can fail, for example when the later model is a specialist or when the data shift, and Gupta et al. (2024) extend this analysis to uncertainty at the level of tokens. Routers choose among several models to balance cost and quality (Ong et al., 2024). Learning to defer trains a classifier together with a rule for passing inputs to a human expert (Madras et al., 2018; Mozannar & Sontag, 2020). Many applications of typed decision models follow the same pattern. A typed decision model handles most inputs, and a reasoning LLM handles the rest (Wu & Lim, 2026b; Li et al., 2026i).

2.9 Fast and slow thinking

The distinction between fast and slow thinking (Kahneman, 2011) has been used before in AI to motivate systems that combine quick pattern recognition with slower reasoning (Booch et al., 2021). For typed decision models the analogy describes a division of labour rather than a specific mechanism. Some recent work takes it further. One study adds limited extra computation inside a single decision (Gan et al., 2026), and another asks when a fast model should stop and pass the problem to a slower one (Bailo, 2026).

3 Formal Description of Typed Decision Models

This section defines typed decision models as an *interface*: a class of models characterized by what they receive and what they return, not by how they compute it. We then analyze the most common way of using them, namely combining them with a fallback model.

3.1 Typed questions

Definition 1 (Typed question)**.** *A* typed question *is a tuple* $q = (\text{type}, \phi, \mathcal{C})$*. Here* $\text{type} \in \{\textsc{Choice}, \textsc{Score}, \textsc{Noul}\}$*, ϕ is the question in natural language, and $\mathcal{C}$ is the* answer specification*:*

- *for* CHOICE*, an ordered list of options $o_1, \ldots, o_n$, each with a name and an optional written definition;*
- *for* SCORE*, an ordered list of levels $0, \ldots, K-1$, each with a label and an optional description;*

Table 1: Earlier lines of work that used language models for decisions, and what typed decision models add.

Line of work	What it provided	What typed decision models add
Label probabilities (Brown et al., 2020; Zhao et al., 2021)	probabilities over label words from one forward pass	questions with declared answer types; many questions at once
LLM-as-a-Judge (Zheng et al., 2023; Liu et al., 2023)	evaluation of open-ended text	no explanation needed; much lower latency
Reward models and verifiers (Ouyang et al., 2022; Cobbe et al., 2021)	scores for quality or correctness	arbitrary questions at inference time instead of one fixed criterion
Safety classifiers (Inan et al., 2023)	safety verdicts over fixed categories	categories defined at inference time
Tool and action selection (Schick et al., 2023)	tool calls written as text	direct choice from a list of actions
Constrained generation (Willard & Louf, 2023)	outputs that follow a schema	no token-by-token generation; probabilities for all answers
Selective prediction (Geifman & El-Yaniv, 2017; Angelopoulos & Bates, 2021)	abstention on uncertain inputs	probabilities and confidence returned with the answer
Cascades and routers (Chen et al., 2023; Ong et al., 2024)	passing hard inputs to stronger models	a general-purpose first stage
Zero-shot encoders (Laurer et al., 2023; Zaratiana et al., 2024)	labels or types given at inference time	probabilities over arbitrary typed questions

- *for* NOUL*, an optional description of when the statement counts as true.*

The answer space $\mathcal{T}(q)$ *is* $\{o_1, \ldots, o_n\}$*,* $\{0, \ldots, K-1\}$*, or* $\{\textsc{true}, \textsc{false}\}$*, respectively.*

The answer specification is part of the question, because names, definitions, and the order of options all affect the output (Sections 9.1 and 9.2). Two questions with the same text and the same answer space can therefore lead to different answers. The answer space is also a *closed-world* assumption: it may contain no correct answer, or more than one. Users can add an explicit option such as "none of the above", but this does not guarantee that the model uses it (Section 9.5).

3.2 Typed decision models

Definition 2 (Typed decision model)**.** *Given a state* s *and typed questions* $q_1, \ldots, q_m$*, a* typed decision model *returns, for each question, an output whose form depends on the question type. Let* $p_j = p_\theta(\cdot \mid s, q_j)$ *denote a probability distribution over* $\mathcal{T}(q_j)$*. Then*

$$M_\theta(s, q_j) = \begin{cases} (\hat{y}_j,\ p_j,\ \kappa_j), & \text{if } q_j \text{ is a } \textsc{Choice} \text{ question, with } \hat{y}_j \in \mathcal{T}(q_j), \\ (\mu_j,\ p_j,\ \kappa_j), & \text{if } q_j \text{ is a } \textsc{Score} \text{ question, with } \mu_j = \sum_{k=0}^{K-1} k\, p_j(k) \in [0, K-1], \\ \pi_j, & \text{if } q_j \text{ is a } \textsc{Noul} \text{ question, with } \pi_j = p_j(\textsc{true}) \in [0,1]. \end{cases} \tag{1}$$

Here $\hat{y}_j$ *is the selected option, usually the most probable one, and* $\kappa_j \in [0,1]$ *is a confidence score reported by the model. A model may also decline to answer a question, which we write as* $M_\theta(s, q_j) = \perp$*.*

This definition follows the interfaces documented by the providers. Jev returns a selected option, probabilities, and a confidence for CHOICE, and a probability-weighted score, probabilities, and a confidence for SCORE. For NOUL, it returns only the probability that the answer is yes, with no separate confidence (TypeSafe AI, 2026b). OpenAI's service has the same structure. It calls the yes-or-no type a *predicate*, its scores "can fall between the labels", and a question may instead receive a *refusal* (OpenAI, 2026). Three points about the definition matter for the rest of the survey.

First, the output of a SCORE question is an expected level μ. It generally lies *between* the levels and is not itself an element of $\mathcal{T}(q)$. Second, NOUL questions have no confidence score; any measure of certainty must

be derived from π. Third, the definition does not prescribe how the outputs are computed. We call a model a *single-pass* typed decision model if it obtains all its outputs from one inference call without autoregressive decoding, as Jev's developers describe their model (Almeida, 2026). Many methods reviewed in Section 4 belong to the broader class but not to this subclass. Examples are methods that average over several option orders (Zhang et al., 2026b), that narrow down a numerical answer in several steps (Ye et al., 2026), or that score each candidate separately. We write *single-pass* explicitly whenever a statement applies only to this subclass.

Well-formed, correct, and calibrated. The outputs in Eq. (1) are always well-formed, that is, they lie in the declared answer space (or are an explicit refusal). Its developers describe this property as being unable to "hallucinate" (Almeida, 2026). Well-formedness, however, says nothing about correctness. A well-formed answer can be wrong, and it can follow the name of an option rather than its definition (Sun et al., 2026; Azizi et al., 2026a). What counts as correct depends on the task. For many classification tasks there is a single reference answer $y^\star(s)$, namely the option whose definition the state satisfies. For subjective tasks, such as preference judgements, rubric ratings, or annotation with disagreement among annotators, the target is better described by a distribution $P^\star(\cdot \mid s, q)$. In that case, p_θ should be compared with $P^\star$ rather than with a single label. Calibration is a third, separate property: the probabilities agree with observed frequencies. It is an empirical property that must be measured, and it is not part of the definition of the class.

Questions are evaluated separately. Jev evaluates each question "in parallel and in isolation" (TypeSafe AI, 2026b). We formalize this as *separate evaluation*: the output for q_j does not depend on which other questions are asked in the same request. This is a property of the interface, not a statement about probabilistic independence. The models do not return a joint distribution over the answers to several questions. Separate evaluation makes answers easy to combine and reuse. However, nothing in the interface enforces logical constraints across questions, such as $P(X) + P(\neg X) = 1$. Several studies indeed find such constraints violated (Section 9.4).

Answer probability, reported confidence, and routing score. Three quantities should be kept apart. The *answer probabilities* p_j form a distribution over the answer space. The *reported confidence* κ_j is a number supplied by the provider. Its definition is not documented: Jev's materials do not state how it is computed (TypeSafe AI, 2026b), and OpenAI recommends choosing thresholds from labeled examples of the target task rather than reading the confidence as a probability of correctness (OpenAI, 2026). Finally, a *routing score* r_j is any number that a system uses to decide whether to act on an answer. It may be the reported confidence, the largest answer probability $\max_y p_j(y)$, or, for a NOUL question, the certainty $\max\{\pi_j, 1-\pi_j\}$. The routing score is chosen by the user of the model, and it need not be calibrated. Studies that examine confidence separately from answer probabilities (Shankaranarayana et al., 2026; Porcedda, 2026) are therefore studying different objects.

3.3 Combining a typed decision model with a fallback

In practice, a typed decision model is rarely used alone. In the most common design, a routing score r and a threshold τ decide whether the model's answer is used. If $r \geq \tau$, the answer is used. Otherwise, the input is passed to a fallback F, such as a reasoning LLM or a human (Li et al., 2026i; Wu & Lim, 2026b; Bailo, 2026). Let $\hat{y}$ denote the model's answer; for a NOUL question, $\hat{y} = \text{TRUE}$ if $\pi \geq 1/2$. Let $A = \{r < \tau\}$ be the event that an input is passed to the fallback. Let $u(s, y)$ be the utility of answer y for state s. Let c_M and c_F be the costs of calling the typed decision model and the fallback, expressed in the same units as u; for example, a monetary cost can be converted with a trade-off factor. The expected utility of the combined system is

$$U(\tau) = \mathbb{E}\Big[\mathbf{1}_{\neg A}\, u(s, \hat{y}) + \mathbf{1}_A \left(u(s, F(s)) - c_F\right)\Big] - c_M. \tag{2}$$

The fallback cost is paid only for inputs that are passed on.

Proposition 1 (Value of the fallback)**.** *Consider a task with a single reference answer $y^\star(s)$ and utility $u(s, y) = \mathbf{1}\{y = y^\star(s)\}$. Let $C_M = \{\hat{y} = y^\star\}$ and $C_F = \{F(s) = y^\star\}$ be the events that the typed decision*

model and the fallback are correct. Compared with always using $\hat{y}$, the combined system changes the expected utility by

$$\Delta(\tau) \;=\; \Pr\left(A \cap \neg C_M \cap C_F\right) \;-\; \Pr\left(A \cap C_M \cap \neg C_F\right) \;-\; c_F \Pr(A). \tag{3}$$

Hence $\Delta(\tau) > 0$ if and only if $\Pr(A \cap \neg C_M \cap C_F) > \Pr(A \cap C_M \cap \neg C_F) + c_F \Pr(A)$. In particular, $\Delta(\tau) \leq \Pr(A \cap \neg C_M \cap C_F)$.

Proof. On $\neg A$, both systems use $\hat{y}$. On A, the combined system obtains $\mathbf{1}\{C_F\} - c_F$ instead of $\mathbf{1}\{C_M\}$, and $\mathbf{1}\{C_F\} - \mathbf{1}\{C_M\} = \mathbf{1}\{\neg C_M \cap C_F\} - \mathbf{1}\{C_M \cap \neg C_F\}$. Taking expectations gives Eq. (3), and the remaining statements follow directly. □

The proposition shows that the fallback helps only if, among the inputs passed on, it corrects more of the model's errors than it introduces new ones, by a margin larger than its cost. A necessary condition is that the fallback is right on some inputs where the model is wrong, that is, the two models must make *different errors*. This condition is not sufficient. A fallback that fixes 10% of the passed-on inputs but breaks another 20% makes different errors and still lowers accuracy. If the two models make the same errors, $\Pr(A \cap \neg C_M \cap C_F)$ is close to zero, and no threshold helps. This is what has been observed for rubric-based judging (Section 7.1). The result agrees with earlier analyses of model cascades, which show that deferral based only on the first model's confidence can be insufficient (Jitkrittum et al., 2023; Gupta et al., 2024).

Optimal routing. The proposition suggests the quantity that routing should target. Define the *fallback advantage*

$$d(s) = \mathbb{E}\big[\, u(s, F(s)) - u(s, \hat{y}) \mid s \,\big] - c_F. \tag{4}$$

The routing rule that maximizes Eq. (2) passes an input to the fallback exactly when $d(s) > 0$. A threshold on a routing score r recovers this rule only if r orders inputs in the same way as $-d$. The reported confidence of the typed decision model is only a proxy for d, because it concerns the model's own correctness and ignores where the fallback does better. Learning a routing score that predicts d directly is the problem studied in learning to defer (Madras et al., 2018; Mozannar & Sontag, 2020).

Calibration is not discrimination. A calibrated routing score does not by itself send the model's errors to the fallback. Calibration only requires, for example, that answers with score 0.8 are correct 80% of the time. Consider a model with 80% accuracy that reports a score of 0.8 for every input. This model is perfectly calibrated, but its score does not distinguish correct from incorrect answers: any threshold passes on either no input or every input. What routing needs is *discrimination*, that is, errors must receive lower scores than correct answers. Discrimination is measured by the area under the ROC curve for correctness or by risk–coverage curves (Geifman & El-Yaniv, 2017). Calibration makes the values of the score interpretable as frequencies of correctness. Both properties are useful, but they are different. Three further points follow.

1. **Accuracy alone does not determine usefulness.** A typed decision model can be less accurate than the fallback and still improve the system, as long as its routing score discriminates well and the fallback is much more expensive. This explains the results on judging with a fallback (Li et al., 2026i) and on agents that call a strong model only when needed (Wu & Lim, 2026b). It also explains why recent benchmarks report probability quality and cost along with accuracy (Sharma et al., 2026; JevBench maintainers, 2026).

2. **What matters is behaviour near the threshold, under realistic data.** The combined system depends on how well r separates correct and incorrect answers around τ. This separation can deteriorate on unfamiliar content (Shankaranarayana et al., 2026) or when the set of options changes (Lu & Wu, 2026). A threshold chosen on one dataset may then let confident errors through.

3. **The gain is limited by the fallback.** If the fallback is itself cheap and accurate, the typed decision model adds little. One study reports exactly this when a simple generative cascade already routes inputs accurately (Wu & Lim, 2026b).

3.4 When is a single pass enough?

A single-pass typed decision model uses the same amount of computation for every question, regardless of difficulty. A reasoning LLM, in contrast, can spend more computation on harder inputs. Several studies report a pattern that we state as a working hypothesis, not as an established result. The relative performance of single-pass models tends to be weakest when the answer must be worked out in several dependent steps. Examples include multi-step arithmetic (Cheng et al., 2026b; Li et al., 2026g; Zhong et al., 2026), predicting consequences that are not stated in the input (Yang et al., 2026b), and graph structure (Yang et al., 2026a). On tasks where the needed evidence is stated in the input, such as classification, extraction, or rubric matching, results are more often competitive with generative LLMs, but they are mixed. Typed decision models still trail supervised models on several such tasks (Xu & Mei, 2026; Konak, 2026) and trail the best LLMs on many annotation tasks (Ibrahim & Zaki, 2026). Having the evidence in the input is therefore not a sufficient condition for good performance. We refer to the hypothesized pattern as the *evidence–computation boundary*. It agrees with the finding that LLMs answering directly, without explicit reasoning, combine sequential evidence unevenly (Xue et al., 2026), and with the finding that strict output formats can harm reasoning (Tam et al., 2024). The boundary is not sharp: one broad evaluation reports relatively good results on calculation questions (Deußer et al., 2026), and results on knowledge questions depend on what the model has learned (Duan et al., 2026). Methods that add computation without generating text may shift the boundary. Examples are repeated application of the same layers (Gan et al., 2026), combining several decision paths (Yan, 2026), and letting code compute intermediate results that the model then evaluates (Yang et al., 2026b).

3.5 Roles of decisions

An analysis of 2,170 public software projects that use Jev identified several roles that decisions play (Ling et al., 2026): judging a property of the input (Noul or Choice), scoring on a rubric (Score), selecting the next action of an agent (Choice), filtering content, and selecting a model or tool. Judging and scoring were common in all domains, while the other roles depended on the domain (Ling et al., 2026). Section 7 uses a refined version of these roles.

4 Models

This section reviews the main types of typed decision models: Jev, similar models from other providers, open models designed for typed decisions, general LLMs used as typed decision models, and multimodal models. Table 2 lists representative examples.

4.1 Jev

What is known. According to its developers, Jev does not generate its output token by token. Instead, it produces the answers to all questions in a single parallel step (Almeida, 2026). It is trained with RLCD, which aims at probabilities that reflect true uncertainty. This goal differs from training on human preferences or on verifiable rewards. Jev accepts text and structured data but not images, audio, or video, and allows up to 255 options per question. The developers report a response time of 70 to 500 milliseconds (Almeida, 2026; TypeSafe AI, 2026b). They also report large speed and cost advantages over frontier LLMs. They note, however, that these figures are probably at the high end of what users will observe. The reference answers in their comparison were also produced by two frontier LLMs, which favours those models (Almeida, 2026). The developers further provide a tool that answers the same typed questions with general-purpose LLMs, so that users can compare the two approaches (TypeSafe AI, 2026a). This suggests that the developers themselves regard the question format and the model as separable.

What is not known. The weights, size, architecture, training data, and exact training objective of Jev have not been released (Li & Wagle, 2026). Some studies infer properties from Jev's behaviour. One suggests that Jev reads its answers from a single forward pass over the input and that its ranking of answers changes

little when the list of options is changed (Li & Chen, 2026b). Another suggests that its probabilities leave out an implicit "unknown" outcome (Porcedda, 2026). Such conclusions about a closed model remain hypotheses.

4.2 Similar models from other providers

Other providers have adopted the same design. Cloudflare released two open models, Clef (27B parameters) and Clef-flash (9B), both fine-tuned from existing open LLMs (Cloudflare, 2026). They accept text, images, and video, answer the same kinds of questions as Jev, and return a probability for every allowed answer. Cloudflare also offers fine-tuning of these models with reinforcement learning. OpenAI introduced a service based on one of its own models (OpenAI, 2026). It supports three question types: *predicates* (the probability that a statement is true), *choices* (one option, with a probability for each option and a confidence score), and *scores* (a probability-weighted average over ordered levels). It accepts text and images. Its documentation recommends choosing confidence thresholds from labeled examples of the target task, rather than assuming that the confidence is calibrated (OpenAI, 2026). Since several providers now use the same three question types, the design is becoming a common format. Models from different providers can therefore be compared on the same questions, although they do not yet define all outputs in the same way (Section 10).

4.3 Open models designed for typed decisions

Encoder-based models. Laya is an open model of about 421 million parameters. It combines a ModernBERT encoder (Warner et al., 2024) with a decision head and answers Choice, Score, and yes-or-no questions in one forward pass (ConvAI Innovations, 2026). An independent, preregistered study reproduced its reported accuracy (0.767 versus 0.766). However, it found the model consistently *under*confident: its average confidence was 0.214 lower than its accuracy, contrary to the model description. A single temperature parameter removed most of this miscalibration. The study also found that a confidence threshold fixed in advance did not reach its error target, and that the model transferred poorly to new data (Nandakishore, 2026b). Chinese-Jev uses a small encoder trained on about ten million Chinese examples. Their labels were converted from different annotation formats into distributions over answer options. It reports higher accuracy on Chinese tasks than Jev, much lower latency, and can run on mobile devices (Wang et al., 2026c). A Turkish model scores each option separately, so that its answers do not depend on the order of the options (Teke, 2026b). SBERT2S1 turns biomedical sentence encoders, originally trained for retrieval, into typed decision models. It also releases a biomedical benchmark and a training set of 243 thousand decisions (Deka, 2026). MetaEncoder adapts a decoder into an encoder that embeds the input and the candidates separately. This allows decisions over images and video and over very large candidate sets (Cheng et al., 2026a). These models show that the question format, rather than a particular architecture, is the essential element. A bidirectional encoder with a scoring head is already enough for a typed decision model.

Encoding candidates separately. Encoders allow candidate representations to be computed once and reused. A study of the model Certo shows the cost of this choice (Rajput et al., 2026). When candidates were encoded separately from the input and the rules, the computation became about five times cheaper. However, the recall of rule-dependent decisions dropped from 1.00 to 0.24. Additional training on counterfactual examples restored part of the loss on synthetic rules, but not on real ones. Encoding candidates separately therefore weakens the very property that typed decision models are meant to provide: following rules written by the user (Section 9.1).

Decoder-based models. The this-that-model is a 2B-parameter model. It reads answers from hidden states at fixed positions and needs about 31 milliseconds per decision on a consumer GPU. It performs poorly on multi-step arithmetic, which is consistent with the evidence–computation boundary (Cheng et al., 2026b). Bongard is a 7B model with an encoder–decoder architecture. The encoder reads the input once, and several decoder branches make separate decisions from it. It is trained in three stages, from supervised labels to the outcomes of actions (Ding et al., 2026). SanSi applies the same layers several times before producing a single answer. This adds computation without generating text and improves accuracy (Gan et al., 2026). SearchJev is a typed decision model for search agents. It is trained with soft labels to improve calibration (Cao et al., 2026). Another study restricts attention so that each candidate sees the shared input but not the

other candidates. This makes the answers less sensitive to the order of the candidates without reducing their quality (Amit, 2026).

Domain-specific models. SecJev adapts a small typed decision model to security tasks and reports large gains over general models (Chen et al., 2026). Other models target phone-scam detection (Ren et al., 2026), penetration testing (dos Santos Barbosa, 2026), and ship design (Huo et al., 2026). Very small specialized models can outperform general ones on their own task. For example, a form-filling model with only 706 thousand parameters reports 99.7% accuracy on its own benchmark, compared with 83.6% for Jev, while Jev is better on decisions that require judgement (Cua AI, 2026). The strength of general typed decision models is therefore their breadth, not their accuracy on any single task.

4.4 General LLMs as typed decision models

An important question is whether specially designed models are needed at all. One study argues that "modern LLMs are inherently effective decision models" (Li & Wagle, 2026). It reads decisions from the next-token probabilities that an LLM assigns to numbered option labels. This works for any number of options and for images, without changing the model. Without any training, a 4B model matched openly released models built on the same base model. Fine-tuning helped weaker models and tasks with many options, but gave little benefit for stronger models (Li & Wagle, 2026). Other studies obtain decisions from the first generated token (Dai et al., 2026; Ding, 2026). InnerJev trains a model to place the result of its own reasoning directly into its first output token. Its 27B version reportedly matches Jev on a broad benchmark at about 0.1 seconds per question (Duan et al., 2026). NumericJev extends this approach to numerical answers by narrowing down the range of the answer step by step (Ye et al., 2026). Distillation offers another route. Typed decisions can be distilled from larger models into small task-specific models (Sonthalia et al., 2026; Wu et al., 2026a), and Jev's own decisions can be distilled into compact evaluators (Sadoun et al., 2026).

AnyJev describes the reading of probabilities from an LLM in detail and corrects two known biases without any training (Zhang et al., 2026b). It removes the model's prior preference for certain labels, which it estimates from unlabeled inputs. It also averages over rotated orderings of the options. On two tasks with 20 options, the share of answers that changed when the order was changed fell from 0.33 to between 0.14 and 0.18, and accuracy improved for all eleven tested models. These corrections correspond to earlier methods for calibrating label probabilities and for multiple-choice answering (Zhao et al., 2021; Robinson et al., 2023). Dyad goes further: it adds an encoder that represents each candidate action and scores it against the internal state of the LLM. Trained with reinforcement learning, it outperforms standard training on interactive tasks (Zhan et al., 2026). SubJudge reads continuous scores from the probabilities of digit tokens (Shi et al., 2026). DecSteer adds a small number of parameters to a frozen model and replaces a long chain of reasoning with a short decision (Li & Chen, 2026a). Another study compared an open model with Jev on 256 public questions with known Jev answers. The open model answered 231 correctly and Jev 238, a difference that is not statistically significant ($p = 0.21$) (bradAGI, 2026). The sample is small, but the comparison uses the same questions for both models and agrees with the academic studies above.

An early review also found no independent accuracy advantage of typed answers over label probabilities (Tang & Zheng, 2026). Together, these results suggest that Jev's main contribution may be a well-designed service rather than an ability that general LLMs lack. Whether its training method gives better calibration than these alternatives is still unknown.

4.5 Open reproductions

Besides the models described in academic papers, many open reproductions of Jev's design have been released. A public leaderboard lists about seventy (apolinario, 2026), and two community lists describe many more (awesome-jev contributors, 2026; heyjunpenn and contributors, 2026). They follow four approaches, which mirror those in the academic literature:

- reading answer probabilities from open LLMs without training (Zhang et al., 2026b; bradAGI, 2026);

- fine-tuning open LLMs of 0.6B to 27B parameters (jaredpalmer, 2026; TokenRhythm, 2026; Bespoke Labs, 2026; denis-pplx, 2026; Contrastive-LM, 2026);
- training small encoders or other compact models, down to specialists with fewer than one million parameters (TianyuCodings, 2026; wfzyx, 2026; Manavarya09, 2026; Cua AI, 2026);
- extending the design to images and other inputs (Xiaooolong, 2026; Cloudflare, 2026).

Some of these models use diffusion language models (Nie et al., 2025), which compute all output positions in parallel and are therefore well suited to reading answer probabilities (heyjunpenn and contributors, 2026). On this leaderboard, the best open reproduction scores about 1.5 points below Jev (apolinario, 2026). Small open models generally score below Jev on broad benchmarks, while specialized models can outperform it on their own tasks (Bespoke Labs, 2026; Cua AI, 2026). Independent evaluations can change the picture. One open model performed close to chance as a judge on RM-Bench and JudgeBench, although its confidence did identify its own errors and its answers rarely changed when the order of the options changed (Nandakishore, 2026a). The quality of open reproductions therefore varies widely, and comparisons should rely on shared benchmarks (Section 8).

4.6 Multimodal models

Jev accepts only text, but the design extends naturally to other inputs, and the models by Cloudflare and OpenAI accept images (Cloudflare, 2026; OpenAI, 2026). PixelJev chooses among candidate labels for an image. Adapting it with a few examples improves accuracy considerably, but does not ensure calibrated probabilities (Zhou et al., 2026a). Visual Jev encodes an image once and answers several questions about it (Yu & Yao, 2026). FastJEV removes redundant computation when scoring candidates for images (Ma et al., 2026a). OmniMed-Jev returns calibrated probabilities over candidate medical diagnoses (Tang & Chen, 2026). NavJev applies typed decisions to navigation guided by vision and language (Sheng et al., 2026). MetaEncoder supports both images and video (Cheng et al., 2026a). One study first describes video frames in text and then asks Jev about them, with mixed results (Qin et al., 2026).

5 Training

5.1 Training objectives

Let $\mathcal{T}$ be the answer type of a question, $p_\theta(\cdot \mid s, q)$ the model's distribution over answers, and $y^\star$ the observed outcome. Training objectives differ mainly in the kind of supervision they use.

Supervised labels. With one correct label per example, the standard objective is the cross-entropy $-\log p_\theta(y^\star \mid s, q)$. This is the logarithmic scoring rule, which is strictly proper: its expected value is minimized only by the true conditional distribution of the labels (Gneiting & Raftery, 2007). This is a property of the population objective. It does not guarantee that a finite model trained with cross-entropy is calibrated, which is why calibration after training is often needed (Section 5.5). When the labels are themselves distributions, for example from several annotators or from a teacher model, training against these soft labels keeps the uncertainty in the data. SearchJev uses such soft labels and reduces calibration error compared with single labels (Cao et al., 2026). Chinese-Jev converts several types of annotation, such as single labels, multiple labels, rankings, and scores, into distributions over answer options (Wang et al., 2026c).

Objectives for many options. When the number of options is large or varies, LLM-as-Jev uses a loss over the whole list of options. The loss is factorized along a tree of option labels, so that any number of options can be handled with a fixed vocabulary (Li & Wagle, 2026).

Reinforcement learning for calibrated decisions. TypeSafe AI describes RLCD only briefly. According to the developers, the probabilities are "optimized against outcomes to reflect uncertainty" (Almeida, 2026; TypeSafe AI, 2026b). A natural interpretation is reinforcement learning with a reward given by a proper

Table 2: Representative typed decision models. "Open" indicates released weights; "method" indicates a technique that applies to existing open LLMs. Figures are as reported by the cited sources.

Model	Open	Design	Notes
Jev	no	not disclosed; parallel output; RLCD training	text only; up to 255 options (Almeida, 2026)
Clef / Clef-flash	yes	fine-tuned open LLMs (27B / 9B)	text, images, video (Cloudflare, 2026)
OpenAI service	no	large proprietary LLM	text and images (OpenAI, 2026)
Laya	yes	ModernBERT encoder with decision head (421M)	underconfident in an independent study (ConvAI Innovations, 2026; Nandakishore, 2026b)
Bongard	yes	encoder–decoder (7B) with several decoder branches	trained in three stages (Ding et al., 2026)
this-that-model	yes	2B; reads hidden states	about 31 ms per decision (Cheng et al., 2026b)
Chinese-Jev	–	small encoder	Chinese; runs on mobile devices (Wang et al., 2026c)
SanSi	–	repeated layers before one answer	extra computation without text (Gan et al., 2026)
MetaEncoder	–	separate encoding of input and candidates	images, video, large candidate sets (Cheng et al., 2026a)
LLM-as-Jev	method	next-token probabilities of option labels	with or without fine-tuning (Li & Wagle, 2026)
AnyJev	method	label-prior and option-order correction	no training (Zhang et al., 2026b)
Dyad	method	candidate encoder scored against LLM state	reinforcement learning; agents (Zhan et al., 2026)
InnerJev	–	reasoning distilled into first token	4B and 27B (Duan et al., 2026)
PixelJev	–	multimodal LLM choosing among labels	images (Zhou et al., 2026a)

scoring rule, for example

$$R_{\log}(p, y^\star) = \log p(y^\star), \qquad R_{\text{Brier}}(p, y^\star) = -\sum_{y \in \mathcal{T}} \big(p(y) - \mathbf{1}\{y = y^\star\}\big)^2. \tag{5}$$

Both rewards are maximized in expectation only when p equals the true distribution of outcomes (Brier, 1950; Gneiting & Raftery, 2007). Note that, for a single observed outcome, $R_{\log}$ is exactly the negative of the cross-entropy loss. The incentive to report honest probabilities therefore does not come from using reinforcement learning instead of supervised learning; it comes from the scoring rule, and both approaches share it. The differences lie elsewhere. In reinforcement learning, the outcome $y^\star$ can be observed only after acting, such as whether a support ticket was resolved, and need not be a human label. The reward can also be assigned to a sampled output, such as a rationale followed by a distribution, through which no gradient can be passed directly. By contrast, reinforcement learning with verifiable rewards (DeepSeek-AI et al., 2025) rewards only whether the chosen answer is correct. This reward is not a proper scoring rule for the reported probabilities, and it pushes the probability of the chosen answer toward one. Similar rewards were proposed earlier to train generative LLMs to state calibrated confidence alongside their answers (Damani et al., 2026; Bani-Harouni et al., 2026).

OpenJev-RLCD is the first public implementation of this idea (Gao & Wang, 2026). The model first writes a short rationale and then gives a distribution over answers, which is scored with a proper scoring rule. The authors show that scoring several samples encourages diverse reasoning, which rewards for correct answers alone do not. They also find that direct optimization is unstable and requires two stages: first calibration, then reinforcement learning. When uncertainty comes from disagreement between annotators, the method cannot outperform cross-entropy training. On small models, it matches supervised fine-tuning in accuracy and improves selective prediction. Because it generates a rationale, OpenJev-RLCD demonstrates the training objective but not Jev's single-pass inference.

Objectives for robustness. Some objectives aim at robustness rather than accuracy. PACT trains on pairs of inputs that differ in one fact that changes the correct answer (Lin, 2026). Its loss includes terms that encourage consistent answers under different option orders, that require the answer to depend on the relevant evidence, and that penalize large errors on ordered scales. It keeps the accuracy of cross-entropy training while reducing sensitivity to option order and halving the variance across training runs. Architectural changes that isolate candidates from each other also reduce order sensitivity (Amit, 2026). Training on counterfactual examples partly restores the rule sensitivity lost when candidates are encoded separately (Rajput et al., 2026).

Is reinforcement learning necessary? Direct comparisons are still rare. When biomedical sentence encoders were turned into typed decision models, plain cross-entropy outperformed a public RLCD implementation by 2.5 to 3.0 points, and temperature scaling calibrated models trained with either objective (Deka, 2026). OpenJev-RLCD also finds reinforcement learning comparable to supervised fine-tuning in accuracy, with an advantage mainly in selective prediction (Gao & Wang, 2026). Some providers nevertheless offer reinforcement learning to adapt their models to new tasks (Cloudflare, 2026).

Training in stages. Bongard is trained in three stages: first on supervised labels, then on relations between concepts, and finally on the outcomes of actions (Ding et al., 2026). Reinforcement learning has also been studied in the other direction, with Jev guiding a learning agent. Jev can serve as a reference policy or guide exploration without fine-tuning (Ma et al., 2026c), and typed decision models have been trained with verifiable rewards to choose examples for in-context learning (Wen et al., 2026).

5.2 Training data

Open typed decision models need training data in the typed question format. One public dataset converts hundreds of existing classification and multiple-choice datasets into this format, with over half a million examples. It shuffles the options of each example so that the position of the correct answer carries no information (sileod, 2026). Chinese-Jev builds a similar collection of ten million examples (Wang et al., 2026c), and SBERT2S1 a biomedical collection of 243 thousand decisions (Deka, 2026). Many open models are trained on labels produced by a larger LLM. Overlap between training data and benchmarks is a related risk, and some benchmarks explicitly warn that they cannot rule it out (hotchpotch, 2026).

5.3 Preserving general abilities

Turning a general LLM into a typed decision model may harm its ability to generate text. LLM-as-Jev adds a penalty that keeps the fine-tuned model close to the original model. With this penalty, the model keeps its conversational ability, and low-rank fine-tuning works best for strong models (Li & Wagle, 2026). This follows the same idea as the penalty used in reinforcement learning from human feedback (Ouyang et al., 2022): the decision ability should be a small change to a strong model, not a replacement of it.

5.4 Adaptation and distillation

Adaptation to new tasks. For image classification, adapting a model with 64 examples per task greatly improves accuracy. It does not, however, ensure calibrated probabilities, and larger models without adaptation sometimes perform better than smaller adapted ones (Zhou et al., 2026a). Small models adapted to one domain can outperform much larger general models in that domain (Chen et al., 2026; Wang et al., 2026c).

Distillation. Because Jev returns probabilities, it is a convenient teacher. Distilling its decisions into small evaluators for chess and retrieval works best when Jev's soft labels are combined with labels from LLMs (Sadoun et al., 2026). Conversely, generative models can be distilled into small routers (Wu et al., 2026a) or into small task-specific models (Sonthalia et al., 2026).

5.5 Calibration after training

Even models trained for calibration usually need adjustment before use. Temperature scaling and related methods are the standard tools (Guo et al., 2017). Adjusting the decision threshold fixes a serious failure in

Table 3: Training objectives for typed decision models.

Objective	Supervision	Representative work
Cross-entropy	one label per example	(ConvAI Innovations, 2026; Wang et al., 2026c)
Cross-entropy with soft labels	distributions from annotators or teachers	(Cao et al., 2026; Sadoun et al., 2026)
Loss over the option list	labels with varying option sets	(Li & Wagle, 2026)
Reinforcement learning with scoring rules	observed outcomes	(Almeida, 2026; Gao & Wang, 2026; Cloudflare, 2026)
Contrastive training	pairs that differ in one fact	(Lin, 2026)
Reinforcement learning with verifiable rewards	correctness	(Wen et al., 2026)
Training in stages	labels, relations, outcomes	(Ding et al., 2026)
Calibration after training	held-out labels or a stronger model	(Guo et al., 2017; Zhong et al., 2026; Li, 2026a)

which Jev almost never chooses "none of the above" when the correct answer is missing (Zhong et al., 2026). Calibration must be repeated when the set of options or the input distribution changes (Lu & Wu, 2026). A small number of decisions by a stronger model can also be used to calibrate a large number of cheaper decisions (Li, 2026a). Finally, the stability of answers under small changes to the option list can predict accuracy without labels (Li & Chen, 2026b).

5.6 Summary

Table 3 compares the training objectives. An important question remains open. It is not known whether reinforcement learning with proper scoring rules gives better calibration than training on soft labels followed by calibration, when both use the same data.

6 Efficiency

Lower latency and cost are the most consistently reported advantages of typed decision models (Section 10). This section explains where these gains come from, relates them to earlier work on efficient inference, and reviews methods that make typed decisions faster still.

6.1 Where the savings come from

Consider an input of L_s tokens and m questions whose text and options together contain L_q tokens. A generative model first processes the whole input in one parallel step, the *prefill*. It then generates N output tokens one at a time, often a rationale followed by an answer. Its running time is roughly

$$T_{\text{gen}} \approx t_{\text{pre}}(L_s + L_q) + N\, t_{\text{dec}}, \tag{6}$$

where $t_{\text{pre}}(\cdot)$ is the prefill time and t_{dec} is the time to generate one token. Generating a token is much slower per token than prefilling, because tokens must be produced one after another and each step is limited by memory access. Reasoning models also make N large. A *single-pass* typed decision model (Section 3) obtains all answer probabilities from one shared computation over the input and the questions, without decoding:

$$T_{\text{typed}} \approx t_{\text{pre}}(L_s + L_q). \tag{7}$$

When several questions refer to the same input, the input needs to be processed only once. The speedup is roughly $T_{\text{gen}}/T_{\text{typed}} \approx 1 + N t_{\text{dec}}/t_{\text{pre}}(L_s + L_q)$. It is therefore largest when a generative model would

write a long rationale and when many questions share one input. Conversely, a long input remains costly for both kinds of models. This estimate applies only to the single-pass subclass. Other typed decision models need more computation: averaging over R option orders requires about R such passes (Zhang et al., 2026b), scoring each of n candidates separately requires up to n passes unless their computation is shared, and step-by-step methods for numerical answers require one pass per step (Ye et al., 2026). Their cost lies between the single-pass case and generation, depending on these factors.

6.2 Earlier work on efficient inference

Each source of savings builds on earlier work. Non-autoregressive models generate all output positions at once (Gu et al., 2018). Diffusion language models also compute all positions in parallel (Nie et al., 2025), and speculative decoding reduces the number of sequential steps in generation (Leviathan et al., 2023). Caching the computation for a shared prefix lets many requests reuse it (Kwon et al., 2023; Zheng et al., 2024b). Reading a relevance decision from the probability of a single token was already used for reranking (Nogueira et al., 2020), and reading a ranking from the first generated token made listwise reranking much faster (Reddy et al., 2024), and multiple-choice answering by option letters made such single-token answers reliable (Robinson et al., 2023). Encoder architectures trade accuracy against cost. Cross-encoders process each input–candidate pair together (Devlin et al., 2019). Bi-encoders encode candidates separately, so that their representations can be stored and reused (Reimers & Gurevych, 2019; Zhang et al., 2025). Late-interaction models lie between the two (Khattab & Zaharia, 2020). Knowledge distillation (Hinton et al., 2015) and quantization (Dettmers et al., 2022; Frantar et al., 2023) make models smaller. Adaptive computation spends more effort only where it is needed, through early exiting (Graves, 2016; Xin et al., 2020; Schuster et al., 2022) or through cascades (Chen et al., 2023). Typed decision models combine these ideas in a particular way: by restricting the output to a fixed set of answers, they avoid text generation altogether.

6.3 Sources of speed

Avoiding text generation. The central idea is to read the decision directly from the model's computation over the input, without generating text. Jev is described as producing its output in parallel rather than token by token (Almeida, 2026). Open models read answers from hidden states at fixed positions (Cheng et al., 2026b), from the next-token probabilities of option labels (Li & Wagle, 2026; Zhang et al., 2026b; Dai et al., 2026; Ding, 2026), or from the probabilities of digit tokens for continuous scores (Shi et al., 2026). Numerical answers can be found by narrowing down their range step by step (Ye et al., 2026). When a task benefits from reasoning, the result of the reasoning can be distilled into the single-pass answer. This yields models that answer in about 0.1 seconds (Duan et al., 2026), or that replace thousands of reasoning tokens with a short decision (Li & Chen, 2026a). In search agents, typed decisions were about five times faster than decisions by generative models of similar size (Cao et al., 2026).

Sharing the input across questions. Questions about the same input can share its computation. Visual Jev encodes an image once and answers several questions about it, which is 8.9 times faster when an image has many questions (Yu & Yao, 2026). In behaviour-tree agents, candidates share the cached computation of the input and are scored two to four times faster (Wake & Wagle, 2026). Bongard reads the input once and makes several decisions from it with separate decoder branches (Ding et al., 2026). FastJEV removes redundant computation when scoring candidates for images. It reduces the computation by 44 to 46% while keeping 94 to 98% of the task score (Ma et al., 2026a).

Encoding candidates. How candidates are encoded determines what can be reused. Encoding each candidate together with the input and the rules is accurate but must be repeated for every input. Encoding candidates separately allows reuse and is about five times cheaper, but it loses most of the sensitivity to rules given in the input (Rajput et al., 2026). Attention patterns that let each candidate see the input but not the other candidates allow parallel scoring and reduce sensitivity to option order (Amit, 2026). Encoding the input and candidates separately also allows decisions over millions of candidates (Cheng et al., 2026a).

Table 4: Reported efficiency of typed decision models and related techniques. Figures are as reported by the cited sources. The baselines differ across studies, so the figures are not directly comparable.

Model or method	Technique	Reported efficiency
Jev (Almeida, 2026)	parallel output	70–500 ms per request
this-that-model (Cheng et al., 2026b)	answers read from hidden states (2B)	about 31 ms, compared with about 8.8 s for large LLMs
SearchJev (Cao et al., 2026)	single-pass scoring of options	5.2–5.3× faster than generative models of similar size
Chinese-Jev (Wang et al., 2026c)	small encoder; 8-bit weights	20.3× faster than its baseline
InnerJev (Duan et al., 2026)	reasoning distilled into the first token	about 0.1 s per question
SubJudge (Shi et al., 2026)	scores read from digit tokens	1.29–261× faster than a 9B reasoning model
DecSteer (Li & Chen, 2026a)	small added decision module	3,685 reasoning tokens replaced by 6 tokens
Visual Jev (Yu & Yao, 2026)	one image encoding for many questions	8.9× faster
SharedKV-BT (Wake & Wagle, 2026)	shared input computation across candidates	2–4× faster
FastJEV (Ma et al., 2026a)	removal of redundant computation	44–46% less computation
Certo (Rajput et al., 2026)	candidates encoded separately	about 5× cheaper, but less sensitive to rules
AnyJev (Zhang et al., 2026b)	early stopping over option orders	2.2× more decisions per second
JEVDB (Wang et al., 2026d)	filtering before any model call	87.4% of candidates removed
REFLEX (Wu & Lim, 2026b)	passing only uncertain inputs on	72.7% fewer calls to a strong model

Small models. Encoder models with a few hundred million parameters answer in tens of milliseconds (ConvAI Innovations, 2026; Wang et al., 2026c; Teke, 2026b). A Chinese encoder model reports a 20-fold speedup over its baseline, about 15 milliseconds per example, and use on mobile devices with 8-bit weights (Wang et al., 2026c). A 2B model needs about 31 milliseconds per decision on a consumer GPU, compared with several seconds for large commercial LLMs (Cheng et al., 2026b). Specialized models can be smaller still (Cua AI, 2026), and the decisions of a large typed decision model can be distilled into small ones (Sadoun et al., 2026).

Spending extra computation selectively. A single-pass model uses a fixed amount of computation for every question, so any extra computation should go where it helps most. The stability of answers under small changes to the option list predicts accuracy without labels. Using it to decide which inputs receive extra passes works better than giving every input the same number of passes (Li & Chen, 2026b). Averaging over different option orders improves accuracy. Applying it only to low-confidence inputs keeps most of this gain at a fraction of the cost (Provatas & Georgakopoulos-Soares, 2026), and stopping early once the answer is stable roughly doubles throughput (Zhang et al., 2026b). Conversely, some models deliberately spend more computation per decision by repeating layers (Gan et al., 2026). At the level of whole systems, passing only uncertain inputs to expensive models removes most calls to those models (Wu & Lim, 2026b; Zhou et al., 2026b). In database queries, simple filters can remove most candidates before any model is called (Wang et al., 2026d).

6.4 Trade-offs

These gains come at a price. Several of the failure modes in Section 9 are side effects of efficiency choices.

- **Several inputs in one request.** Asking many questions about one input is safe. Placing several inputs into one request, however, lets them influence each other's answers and makes rankings worse (yodablocks, 2026; anessbelbati, 2026).
- **Reuse versus rule-following.** Encoding candidates separately makes them reusable but weakens sensitivity to rules in the input (Rajput et al., 2026).
- **Robustness requires extra passes.** Removing the effect of option order requires averaging over several orders. Applying this only to uncertain inputs, or stopping early, recovers most of the benefit at lower cost (Zhang et al., 2026b; Provatas & Georgakopoulos-Soares, 2026).
- **More options cost more.** Decoder-based models become slower as options or questions are added (Sharma et al., 2026). Long inputs also remain expensive, and with discounted batch processing of generative models the cost advantage can disappear (Denney & DiGiuseppe, 2026).
- **Fixed computation may limit what can be decided.** Single-pass models appear to struggle with multi-step reasoning (Section 9.6). Such inputs must be passed to a stronger model, and the benefit of doing so is limited by how often the two models make different errors (Proposition 1).

7 Applications

We group applications by the role that the decision plays (Section 3). Table 5 gives an overview.

7.1 Judging and evaluation

Judging is the most direct test of whether a typed decision model can replace a generative one. LLM-as-a-Judge is already a decision task, and its agreement with humans is well studied (Zheng et al., 2023).

Judging with a fallback. One study compares Jev with sixteen generative judges and reward models, using blind human judgements as the reference (Li et al., 2026i). Jev comes within three points of a GPT-6 judge when the verdict "can be read off the text". It costs 0.36% as much and has a median latency of 0.15 seconds. It falls behind when the verdict must be worked out, as in mathematics, code, and logic. Its confidence identifies these cases. With a threshold fixed in advance, accepting confident verdicts and passing the rest to GPT-6 is 0.9 points more accurate than GPT-6 alone, at 41% of its cost. In a live test specified in advance, this combination matched the accuracy of GPT-6 exactly. It worked less well on pairs designed to mislead through style and on texts without a reference answer.

Rubric-based judging and shared errors. Rao & Callison-Burch (2026) compare Jev with three fast LLM judges on nine human-labeled datasets from seven benchmarks. All judges receive the same criteria. The LLM judges cost 16 to 325 times more and are 28 to 350 times slower. Yet Jev's accuracy differs significantly from theirs in at most 8 of 27 comparisons. Jev is better on yes-or-no criteria and worse on ordered scales. The key finding is that *the two kinds of judges make the same errors.* When Jev is confidently wrong, about 96% of the LLM verdicts repeat the same wrong answer. As a result, even with the best possible threshold, no combination beats the best single judge by more than 2.7 points. As the authors put it, "calibration can tell a cascade when to defer, but the cascade also needs a fallback that errs elsewhere" (Rao & Callison-Burch, 2026). This matches Proposition 1. Combining models reliably saves cost, but it can improve accuracy only when the fallback corrects errors of the typed decision model, as on questions that require reasoning.

Consistency is not correctness. A small study with five agent responses, each judged 100 times, found that Jev agreed with human pass/fail labels in all 500 cases. Its scores also varied far less across repetitions than those of three generative judges (Shea & Roche, 2026). The sample is too small to generalize, but it illustrates that single-pass models are nearly deterministic. Consistency alone, however, does not imply correctness. An open typed decision model evaluated on public preference benchmarks, including RM-Bench and JudgeBench (Liu et al., 2025; Tan et al., 2025), was very consistent but performed close to chance (Nandakishore, 2026a).

Safety judging. On 5,219 agent trajectories from four security benchmarks, Jev reaches an average F1 of 77.8, compared with 74.1 for the strongest generative judge, at about $0.0002 per judgement. The ranking, however, varies across datasets (Wang & Gao, 2026). Another study tests Jev without training on ten kinds of alignment failure across 44 benchmarks, including sycophancy, jailbreaks, deception, and hallucination (Guo et al., 2026). A single general question reaches a median AUROC of 0.886, outperforms supervised baselines on most benchmarks, and costs 63 times less than LLM judges. Many of these failures can only be judged in relation to something outside the response, such as the user's belief or an injected instruction. The study therefore varies what Jev is asked separately from what it is shown. Other studies warn that high overall accuracy can hide errors concentrated in particular attack types (Liu, 2026b). Another study in the SHADE-Arena environment (Kutasov et al., 2025) shows that the unit of judgement matters (nican2018, 2026). When judging whole agent transcripts, Jev reached an AUC of 0.78, compared with 0.99 for a generative monitor, partly because long transcripts had to be split. When judging each tool call before it was executed, Jev reached 0.97. Combining Jev with a generative monitor detected every attack with far fewer calls to the generative model. However, with thresholds chosen on separate tasks, all Jev-based designs missed an attack disguised as a harmless rule.

Judging in specific domains. Jev has been evaluated as a judge of factual accuracy in radiology reports (Huang et al., 2026), of financial evidence (Xu et al., 2026a), of policy arguments (Loi, 2026), and of video quality (Robitza, 2026). Typed visual judgements have also been used as rewards for training image generators (Li et al., 2026a), which links typed decision models to reward modeling.

7.2 Routing and network control

Routing is the area where low latency matters most, and where the evidence for practical benefit is strongest.

Network control. In one study, Jev extracts four to eight structured fields from natural-language service requests at the network edge, which a scheduler then uses (Li et al., 2026c). Across 8,280 requests, it reduces median decision time by 22.7 to 64.5% compared with the fastest LLM tested. In live operation it completes 91 to 95% of requests, while the LLMs complete fewer than 10%. In the control of 6G radio access networks, Jev meets one-second time limits on nearly all requests, while LLMs meet them only rarely (Li et al., 2026d). Similar studies address antenna selection (Rahimi et al., 2026), 5G network control (Kourtis & Xilouris, 2026), and traffic classification (Xu & Mei, 2026). A review of 139 papers on such decision components in network control finds that most do not measure whether time limits are actually met (Li et al., 2026b).

Selecting models and components. Using Jev to select which LLM should handle each request is reported to reduce the cost of model use by 13 to 21% (Kwartler et al., 2026). Jev has also been used to select among fine-tuned model components without further training (Wang et al., 2026b). Other work uses calibrated confidence to balance accuracy and cost when answering database queries (Kim, 2026), and to decide when model outputs may be trusted in automated decision procedures with statistical guarantees (Besanson, 2026).

7.3 Agent control

The most active area combines a typed decision model with a generative agent. The typed decision model makes frequent, simple choices. The LLM handles planning, text generation, and uncertain cases.

Selective use of a strong model. REFLEX uses Jev for routine decisions and calls a stronger LLM only when confidence is low or text must be generated (Wu & Lim, 2026b). On 100 tasks, it reaches 95% success with 72.7% fewer calls to the strong model. The benefit is smaller when a simple generative cascade already routes accurately. Another study asks directly when a fast model should stop and pass a decision to a reasoning model (Bailo, 2026). A paired comparison of typed decision models in agents finds Jev significantly more accurate than Laya on 9 of 11 types of decisions (Li, 2026b).

Coordination of several agents. S1-MAS assigns coordination decisions in multi-agent systems, such as which task to work on and when to stop, to small typed decision models, and leaves reasoning to LLM agents. It reports large reductions in tokens and latency with equal or better accuracy than earlier multi-agent methods (Zhou et al., 2026b).

Memory. Jev-Mem uses a typed decision model to manage an agent's memory. It reduces the time to build and query the memory and improves task performance (Jiang et al., 2026). Mnemon combines raw records, fast decisions by Jev, and slower planning by an LLM (Wang, 2026). A preregistered study finds that storing past dialogue turns selected by Jev works as well as storing summaries extracted by an LLM, at a small fraction of the cost (Sharma & Lall, 2026).

Robots, interfaces, and games. Jev has been used to execute frequent actions under a slower vision–language planner in mobile interface agents (Zhang, 2026), to decide when a robot can reuse earlier actions (Liu et al., 2026), to make fast decisions in the game StarCraft II (Ma et al., 2026b), and to score candidate actions in behaviour trees (Wake & Wagle, 2026). Other work links task descriptions to sets of possible actions (Su & Huang, 2026) and builds typed decision models for search agents (Cao et al., 2026). In reinforcement learning, Jev can guide an agent without being trained itself (Ma et al., 2026c).

Evaluating versus predicting. One study summarizes a useful division of labour in its title: "code owns the simulation, Jev owns the evaluation" (Yang et al., 2026b). Jev evaluates well the options that are described in the input. It fails, however, when it must predict consequences that are not stated. Code or an LLM should therefore compute such consequences, which Jev then evaluates. Related work studies which decisions a small model should make itself and which it should leave to other components (Fan et al., 2026). Typed decisions have also been combined with fixed rules for checking and approval (dos Santos Barbosa, 2026; Wu et al., 2026b).

7.4 Safety, security, and moderation

Content moderation suits typed decision models well. The decisions are simple, their number is large, and users notice delays. Jev matches specialized moderation systems and improves when similar past cases are added to the input (Mazzoni & Failla, 2026). On hate speech, it comes close to commercial LLMs at a fraction of the cost, although explicit definitions of the categories help little (Paschalides et al., 2026). In security, Jev has been applied to network intrusion detection (Severo et al., 2026), to the classification of software vulnerabilities (Rezvani et al., 2026), to the first screening of anomalies in water distribution networks (Mu et al., 2026), and to the detection of phone scams (Ren et al., 2026). A small model adapted to security tasks has also been proposed (Chen et al., 2026). The security weaknesses of typed decision models themselves are discussed in Section 9.3.

7.5 Database queries and ranking

Typed decisions fit naturally into database queries that contain a semantic condition, such as "does this review complain about delivery?", which must be checked for many rows. JEVDB first removes candidates with standard database operations and simple necessary conditions. It then applies typed decision models to the remaining candidates and passes only unclear cases to generative LLMs (Wang et al., 2026d). In this way, it removes most candidate pairs before any model is called and answers all 21 tested queries fastest, while keeping high accuracy. Calibrated confidence also helps to choose among query plans with different accuracy and cost (Kim, 2026). Further studies suggest caution when the probabilities are used for ranking. Sorting rows by Jev's probability worked for simple topic membership but failed most tests on graded product relevance, and many rows received the same maximal probability (yodablocks, 2026). Placing 40 rows into one request halved the cost but strongly reduced ranking quality. For retrieval, the evidence is mixed. One comparison over 14 datasets found Jev statistically indistinguishable from a commercial reranking model, at less than a fifth of the cost (anessbelbati, 2026). Another study found that reranking with Jev did not outperform embedding-based retrieval (heyjunpenn and contributors, 2026). A probability that is useful for classification is therefore not automatically useful for ranking.

Table 5: Applications of typed decision models, with representative work.

Role	Question type	Representative work
Judging and evaluation	Choice, Score	(Li et al., 2026i; Rao & Callison-Burch, 2026; Wang & Gao, 2026; Guo et al., 2026; Huang et al., 2026; Xu et al., 2026a)
Routing and network control	Choice	(Li et al., 2026c;d; Kwartler et al., 2026; Wang et al., 2026b; Rahimi et al., 2026; Li et al., 2026b)
Agent actions	Choice, Noul	(Wu & Lim, 2026b; Bailo, 2026; Zhang, 2026; Ma et al., 2026b; Su & Huang, 2026; Yang et al., 2026b)
Agent memory	Noul, Score	(Jiang et al., 2026; Wang, 2026; Sharma & Lall, 2026)
Moderation and security	Noul, Choice	(Mazzoni & Failla, 2026; Paschalides et al., 2026; Severo et al., 2026; Mu et al., 2026; Rezvani et al., 2026)
Annotation and measurement	all	(Rafe & Das, 2026a; Denney & DiGiuseppe, 2026; Ibrahim & Zaki, 2026; Zhang et al., 2026c; Lee & Park, 2026)
Database queries	Noul, Score	(Wang et al., 2026d; Kim, 2026)
Coordination of agents	Choice	(Zhou et al., 2026b)
Information gathering	Noul	(Yilmaz et al., 2026; Kang et al., 2026; Yan, 2026)

7.6 Large-scale annotation and measurement

Cheap and calibrated decisions make it practical to label very large text collections. Jev has labeled about 196,000 police reports of traffic crashes (Rafe & Das, 2026a), identified documented pregnancies in millions of such reports (Rafe & Das, 2026d), and, together with a model for tabular data, estimated crash factors that are missing from coded records (Rafe & Das, 2026b). The same authors also compare typed decision models with trained classifiers and LLMs (Rafe & Das, 2026c). In political science, Jev matches LLMs in seven replication studies and is better calibrated, but has no clear cost advantage when the LLMs are used with discounted batch processing (Denney & DiGiuseppe, 2026). Other applications include sentiment analysis (Zhang et al., 2026c), emotion classification (Zhang & Yu, 2026), stance detection (Li et al., 2026h), surveys of cultural values (Asseri & Asseri, 2026), legal documents (Zhang et al., 2026a), medical questions (Madrid-García & Merino-Barbancho, 2026), scientific workflows (Deng et al., 2026), recommendation (Lyu & Xia, 2026), matching database schemas (Li et al., 2026f), tracking student knowledge (Lee & Park, 2026), a speech prosthesis (Cinà, 2026), fault diagnosis in buildings (Jung, 2026), ship design (Huo et al., 2026), materials discovery (Kang et al., 2026), and simulating populations (Li, 2026a).

The largest evaluation in this area follows a standard protocol for LLM annotation on 18 tasks from computational social science (Ibrahim & Zaki, 2026). The typed decision model performs worse than the best LLM on 14 of 15 comparable tasks, with a median gap of 11.6 macro-F1 points, but at a median cost 44 times lower. However, passing its low-confidence items to an LLM matches or exceeds the LLM alone, at a quarter to half of the cost. This is a direct example of Proposition 1. The study follows earlier evaluations of LLMs for text annotation (Ziems et al., 2024), which also warn that annotation errors can bias the conclusions drawn from the labels (Egami et al., 2023; Baumann et al., 2025). Another study suggests that typed decision models can also approximate the aggregate behaviour of people. Given only the two headlines from each of 10,984 randomized A/B tests (Matias et al., 2021), Jev picked the more successful headline in 64.5% of the cases, and in 74.7% when the difference was large (Gaurav-Gosain, 2026). Not all applications succeed. For activity recognition from sensor data, Jev without training performs far worse than supervised models, and adding more features makes it worse (Konak, 2026).

Deciding whether to ask. Several studies treat the decision of *whether to gather more information* as a typed decision. LAVOIR extends Laya to decide when and what to ask, based on the expected value of the information (Yilmaz et al., 2026). CrystalJev uses a similar rule to decide when to run expensive simulations of materials (Kang et al., 2026). JevForest combines several decision paths to decide which features to acquire under a budget (Yan, 2026). These methods are examples of the combined system in Eq. (2), with structured fallbacks.

8 Evaluation

8.1 What should be measured

The answers of a typed decision model are usually used directly by a program rather than read by a person. Accuracy alone is therefore not enough. Following Eq. (2) and the view of Sharma et al. (2026) that these models must be evaluated jointly on rule-following, robustness to wording, quality of probabilities, and the outcomes of the resulting decisions, we distinguish five groups of measures. The first is *answer quality*, measured by accuracy, F1, AUROC for yes-or-no questions, or rank correlation for scores. Because results change with the wording of a question, accuracy should be reported over several paraphrases (Sharma et al., 2026). The second is *probability quality*. It is measured by the expected calibration error (Naeini et al., 2015) and by proper scoring rules (Brier, 1950; Gneiting & Raftery, 2007). Other useful measures are the distance to known true probabilities (Porcedda, 2026) and the consistency of answers to logically related questions (Li et al., 2026e; Joy, 2026; Chen & Li, 2026). The third is the *quality of selective decisions*, measured by the error at a given share of answered inputs and by the cost of the resulting decisions. For example, when errors have unequal costs, acting on the probabilities of an underconfident model can be worse than simply taking its top answer (Sharma et al., 2026). The fourth is *robustness*, measured by how often answers change under rewording, renamed options, added context, or injected text, compared with how often they change when the same request is simply repeated (Hu et al., 2026). The fifth is *efficiency and task success*: latency, the share of decisions made within a time limit (Li et al., 2026d), the cost per correct decision (Li et al., 2026c), and the success of complete tasks that involve many decisions (Duan et al., 2026).

8.2 Benchmarks

Table 6 lists benchmarks designed for or used with typed decision models.

General capability. JEVal contains 11,257 examples from 36 datasets in two languages and ten domains (Duan et al., 2026). It finds that typed decision models perform well when a decision "can be resolved from available evidence" but worse when specialist knowledge or honest uncertainty is needed. They often "identify the most likely outcome while substantially overstating its probability". A broad evaluation on 37 datasets finds Jev ahead of a 27B open LLM on most of them, with well-calibrated probabilities (Deußer et al., 2026). Other comparisons find Jev competitive on knowledge questions but behind on multi-step mathematics (Li et al., 2026g). GraphDecide shows that Jev recognizes whether two nodes are connected but not the broader structure of a graph (Yang et al., 2026a).

Calibration and robustness. TypedBench generates questions with known rules and evaluates commercial and open models on the same items (Sharma et al., 2026). The commercial model follows the given rules, but is sensitive to wording and consistently underconfident. Open decoder models become slower and less accurate as more options are added. Sys1Cal-v1 contains statements whose true probabilities are known, which allows a direct check of calibration (Porcedda, 2026). JevAdvBench contains 812 questions and 9,744 attack variants (Hu et al., 2026). RLCDAlignBench covers ten kinds of alignment failure (Guo et al., 2026). HakemBench is an open Turkish benchmark with 4,275 questions in seven areas. It measures accuracy, calibration, and abstention, and also tests option order, paraphrasing, translation, and substituted names. Its authors note that most of its labels were produced by models (Teke, 2026a). PolicyBench states the decision rule only in the option definitions, and thus tests whether models follow definitions or names (Azizi et al., 2026a).

Other public benchmarks. Several further benchmarks are publicly available (Appendix C). A public leaderboard evaluates models on a fixed set of 37 benchmarks without prompt tuning, and counts missing answers as wrong (apolinario, 2026). Another covers 137 English benchmarks with all three question types, and states that it cannot rule out overlap with training data (hotchpotch, 2026). A third scores every move in a puzzle against an exact solver, which separates the accuracy of single decisions from the completion of whole tasks (Exemocaro, 2026). Further studies publish the raw model outputs of calibration tests (scienthoon, 2026) and use paired designs that change only the language of the input or of the instructions (marcosmartinez,

Table 6: Benchmarks for typed decision models.

Benchmark	Scale	Focus
JEVal (Duan et al., 2026)	11,257 instances, 36 datasets, 10 domains	evidence versus knowledge; combined decisions
TypedBench (Sharma et al., 2026)	7 generators, 9 suites	rule-following, wording, probabilities, cost
Sys1Cal-v1 (Porcedda, 2026)	propositions with known $P(A)$	calibration across question types
JevAdvBench (Hu et al., 2026)	812 questions, 9,744 variants	robustness to attacks
RLCDAlignBench (Guo et al., 2026)	10 failure types, 44 benchmarks	detection of alignment failures
GraphDecide (Yang et al., 2026a)	graph tasks	reasoning about graphs
PolicyBench (Azizi et al., 2026a)	synthetic routing	option definitions versus names
CallScreenBench (Ren et al., 2026)	phone-call transcripts	scam detection
HakemBench (Teke, 2026a)	4,275 questions, 7 tracks	Turkish; accuracy, calibration, abstention
JevBench (JevBench maintainers, 2026)	v1.3.0: 52 systems, 534 decisions	combined score of accuracy, calibration, speed, cost
Decision Index (apolinario, 2026)	37 benchmarks	comparison of open models
S1MB (hotchpotch, 2026)	137 benchmarks	all three question types
System-One Control Bench (Exemocaro, 2026)	100 grid puzzles	single decisions versus whole tasks

2026). These resources are reproducible and independent of the model providers, but they have not been peer reviewed.

Domain benchmarks. Domain-specific evidence comes from benchmarks for scam detection (Ren et al., 2026) and content moderation (Mazzoni & Failla, 2026), and from studies in medicine (Madrid-García & Merino-Barbancho, 2026), law (Zhang et al., 2026a), networking (Li et al., 2026d;c), and political science (Denney & DiGiuseppe, 2026).

8.3 Common problems in current evaluations

Several methodological problems recur in current studies. Some studies address them explicitly (Li, 2026b; Tang & Zheng, 2026).

- **Reference labels from models.** Some comparisons use LLM outputs as the reference (Almeida, 2026), and in robustness studies "most available labels come from the model itself" (Hu et al., 2026). One solution is to compare each answer after an attack with the model's own answer before the attack (Hu et al., 2026).
- **Changing models.** The same request can give different answers, and a commercial model may change between versions (Hu et al., 2026). Studies should report the model version and repeat each request.
- **Benchmarks by model developers.** Benchmarks designed by a model's developer, or tuned on its outputs, should be interpreted with care.
- **How answers are read out.** Comparisons between Jev and open LLMs often differ not only in the model but also in how the answer is read out, for example from option letters, numbers, or stated probabilities. The readout alone can change accuracy and consistency substantially (Li & Wagle, 2026; Li et al., 2026e).
- **How options are presented.** Whether options are shown with their names, their definitions, or both can create or remove biases without any change to the model (Azizi et al., 2026a).
- **Question type and grouping.** The same question asked as Choice or as Noul can give very different probabilities (Zhong et al., 2026; Porcedda, 2026), and placing several inputs into one request changes the answers (yodablocks, 2026).

- **Cost comparisons.** Cost advantages depend on how generative models are priced. With discounted batch processing, the advantage can disappear (Denney & DiGiuseppe, 2026).
- **Small samples.** Some studies report a single run (dos Santos Barbosa, 2026) or a small exploratory sample (Qin et al., 2026). Preregistration (Sharma & Lall, 2026; Li et al., 2026i) and paired comparisons (Li, 2026b) give stronger evidence.

Recommendations. These problems suggest a few principles for future evaluations. Evaluations should test answers under several wordings, option orders, and option names, and should compare these changes with the variation from simply repeating a request. Calibration should be reported separately for each question type and language. When a typed decision model is combined with a fallback, the overlap of their errors should be reported together with accuracy and cost (Proposition 1). Finally, systems that make many decisions in sequence should be evaluated on the success of the whole task, not only on single decisions.

9 Failure Modes

Because the outputs of typed decision models are restricted to the declared answer space (or an explicit refusal), these models avoid a class of errors that affects generative pipelines, such as malformed outputs and answers outside the allowed set. The studies reviewed here show, however, that other kinds of errors remain. Some even become harder to notice, because the answer comes without an explanation (Sharma et al., 2026). We group the failures into six kinds, one per subsection, and summarize them in Table 7.

9.1 Following option names instead of definitions

An option in a CHOICE question has a name and may have a definition. The user's intended rule is stated in the definition, while the name is only a label. One study changed the option names from neutral identifiers to names with meaning, such as "yes" and "no", while keeping the questions and definitions unchanged (Sun et al., 2026). The share of changed decisions rose by up to 70.4 percentage points, and the mean AUC fell from 93.8% to below chance, although every answer still had a valid format. A second study examined four open typed decision models and found that their answers depend "mostly on the labels" (Azizi et al., 2026a). Removing all definitions left the accuracy almost unchanged, and renaming the options to A and B even *increased* it. Importantly, the study traced the problem to how the options are presented to the model rather than to the model itself. Presenting each option as "name: definition" caused the bias, while presenting only the definition removed it, without any change to the model's weights. The problem is the counterpart of the label biases known from earlier work on label probabilities (Zhao et al., 2021; Zheng et al., 2024a). Encoding candidates separately from the input makes it worse, because such models lose most of their sensitivity to the rules given by the user (Rajput et al., 2026). Models should therefore be tested with neutral or contradictory option names before use (Azizi et al., 2026a).

9.2 Sensitivity to context and option order

The answers of typed decision models change with text that a person would consider irrelevant, and with the order of the options. In a study of 6,020 multiple-choice questions related to biosecurity, rotating the options changed the answer for 37.4% of the questions in one subset (Provatas & Georgakopoulos-Soares, 2026). Repeating identical requests showed that this was caused by the order and not by random variation. Averaging over several orders improved accuracy, and doing so only for low-confidence questions kept most of the gain at a fraction of the cost. Averaging over orders is also the main correction used when reading answers from open LLMs (Zhang et al., 2026b). Added context has a similar effect. JevOut constructs short, natural-looking background sentences that are not instructions (Xu et al., 2026b). They change 61.4 to 73.2% of initially correct decisions to a chosen wrong answer, often with high confidence, across four models including Jev. Human annotators judged 91.6% of these additions natural and consistent with the original answer, and the additions also worked on other models. In another study, changing only the presentation of a request raised the rate at which Jev accepted wrong math answers from 1% to 26% (Liu, 2026a). Changing numbers in financial citations revealed a trade-off between catching errors and accepting valid evidence (Xu et al.,

2026a). TypedBench reports sensitivity to the wording of questions (Sharma et al., 2026). JevAdvBench finds that rewording a question changes answers hardly more than repeating it, but that adding a single unverified opinion to the input changes 12.1% of the decisions (Hu et al., 2026). These findings agree: the content of the input matters more than the wording of the question, as another study also observes (Guo et al., 2026).

9.3 Prompt injection and security

A typed decision model can only return an allowed answer. A prompt injection therefore cannot make it produce arbitrary text or call a tool that is not on the list. It can, however, shift probability between the allowed answers. On 510 cases from InjecAgent (Zhan et al., 2024), malicious content shifted the probabilities of actions but rarely made Jev choose the attacker's target (Wu & Lim, 2026a). Adaptive attacks doubled the probability of the target during optimization, and raised the success rate on new requests from 1.8% to 3.5%. Successful attacks occurred mainly where the margin between the top answers was small. In JevAdvBench, the strongest injected instruction changed 10.1% of the decisions, about as many as an added opinion (Hu et al., 2026). Attacks also pushed 38% of confident answers below a confidence threshold of 0.8, which would send them to human review and so overload it. A broader study finds Jev vulnerable to prompt injection and adversarial suffixes (Wang et al., 2026a). It also shows that open models of the same kind can be given hidden behaviours by poisoning their training data, and that private information can be inferred despite the restricted output. The most direct test of typed decision models as safety filters for agents is the *option-channel attack* (Azizi et al., 2026b). On prompt injection, jailbreak, and content screening tasks, typed decision models reached only 36 to 72% accuracy. Irrelevant server logs in the input raised the rate of missed attacks from 0% to 63%, and misleading option names raised it to 93 to 100%. Every tested defence could be defeated, while simple fixed rules solved the same tasks perfectly. The authors conclude that typed decision models should not be the main safeguard for security decisions, although they can reduce the workload of human reviewers. In short, "applications built on RLCD models should therefore treat the state as untrusted, argued input" (Hu et al., 2026). This is the same principle as for indirect prompt injection against generative agents (Perez & Ribeiro, 2022; Greshake et al., 2023).

9.4 Calibration and consistency of probabilities

Calibrated probabilities are the main claim made for Jev, and they have been examined most closely. Studies differ in what they evaluate: some assess the answer probabilities, others the reported confidence, and others a routing score derived from them (Section 3). We note the quantity where it matters.

Average calibration is often good. Several independent studies find Jev's probabilities well calibrated on familiar multiple-choice tasks (Deußer et al., 2026; Denney & DiGiuseppe, 2026; Shankaranarayana et al., 2026). Its confidence also separates tasks it can solve from tasks it cannot (Li et al., 2026i).

The direction of miscalibration varies. TypedBench finds Jev consistently *underconfident* (Sharma et al., 2026). JEVal, in contrast, finds that typed decision models "substantially overstate" the probability of their top answer (Duan et al., 2026). An independent, preregistered study finds the open model Laya consistently underconfident, contrary to its description (Nandakishore, 2026b). A study on biosecurity questions finds good overall calibration (expected calibration error 0.034), but only "once the vendor's uncertainty field is correctly interpreted", and worse calibration on harder tasks (Provatas & Georgakopoulos-Soares, 2026). In computational social science, the confidence of a typed decision model is better calibrated than the stated confidence of most LLMs, yet the model is highly confident while performing near chance on one task (Ibrahim & Zaki, 2026). These studies use different models, question types, and tasks. Their disagreement shows that calibration is not a fixed property of a model, and that the meaning of the returned confidence score must be clearly documented.

Probabilities depend on the question type. Several studies show that the question type changes what a probability means. When the correct numerical answer is missing from the options, a CHOICE question with a "none of the above" option almost never selects it. A NOUL question about each option, in contrast, almost always identifies the missing answer correctly (Zhong et al., 2026) (Section 9.5). Asking about the

same statements with NOUL, CHOICE, and SCORE questions gives systematically different probabilities. CHOICE questions behave as if an implicit "unknown" outcome were left out (Porcedda, 2026). Separate NOUL questions about each label also give higher probabilities than the corresponding CHOICE question (Li et al., 2026e). Further studies confirm this and help to explain the different directions of miscalibration above. On 900 new synthetic questions, CHOICE and SCORE answers were overconfident, while yes-or-no answers were underconfident, so that a single temperature cannot correct both (scienthoon, 2026). Asked 400 times to predict the roll of a fair die as a six-way CHOICE, Jev picked the same face every time, with a mean probability of 0.83 and an accuracy of 19%. A 30% risk stated in a document was returned as 5.3% by a CHOICE question but much more accurately by a NOUL question (Hayashi, 2026). In a simulated survey with 24,596 paired answers, switching from CHOICE to NOUL changed the error 2.3 times more than the entire difference between Jev and a generative LLM (jjd-lab, 2026). Taken together, these results show that a probability from a typed decision model depends on the model, the question type, and the wording. The probabilities of CHOICE questions, in particular, behave more like scores for choosing an option than like calibrated estimates of chance.

Confidence does not indicate missing knowledge. Calibration on average does not show whether low confidence reflects chance or missing knowledge. When given no relevant information, Jev assigned up to 0.80 to a salient option (Shankaranarayana et al., 2026). On news after its training data, its confidence exceeded its accuracy by 0.21 to 0.33, and recalibration on earlier data did not close the gap. Direct NOUL questions, such as whether the evidence is sufficient, were better indicators than the confidence of the answer.

Probabilities are not consistent across questions. Each question is evaluated separately, and nothing in the interface enforces logical constraints across related questions. Related answers therefore need not agree. For a statement X and its negation, Jev's probabilities $P(X)$ and $P(\neg X)$ differ from summing to one by 0.064 on average (Li et al., 2026e). This is much better than reading probabilities from a 27B open LLM (0.293), but about five times larger than the variation between repeated requests. The probabilities of three mutually exclusive labels sum to 1.14 on average. Across 72 thousand classification questions, direct predictions and predictions reconstructed from related questions differ substantially (Joy, 2026). Comparisons of several models on small, fully specified problems find similar inconsistencies (Chen & Li, 2026). One study offers an explanation: in yes-or-no CHOICE questions, Jev behaves as if it leaves out a third "unknown" outcome. Accounting for this outcome raised the median agreement with the true probabilities from 0.771 to 0.978 (Porcedda, 2026).

Rounded probabilities and statistical guarantees. One study reports that many probabilities returned by Jev are rounded to exactly 1.0. In its routing task this was the case for 56.4% of the answers, which limits how low the error of any threshold can be (nikkoxgonzales, 2026). Another reports many rows with the same maximal probability (yodablocks, 2026). The first study also shows what statistical guarantees can achieve. Conformal risk control (Angelopoulos et al., 2024) found a threshold with a guaranteed error of at most 5%. With this threshold, 84.75% of the queries were handled automatically with an error of 2.65%, and the guarantee held on new data. Prediction-powered inference (Angelopoulos et al., 2023) reduced the number of human labels needed to check the system. However, a change in the frequency of classes increased the measured error by a factor of 3.6, and new types of queries broke the guarantee entirely (nikkoxgonzales, 2026).

Calibration does not transfer to new option sets. Thresholds and calibration learned on one set of options often fail when the options or the input distribution change (Lu & Wu, 2026). The ranking of answers, however, can be stable enough under changes to the option list to predict accuracy without labels (Li & Chen, 2026b).

9.5 Missing answers and ordered scales

Missing answers. TypeSafe recommends adding an "other" or "none of the above" option when no listed answer may apply. On paired arithmetic problems, Jev was 99% accurate when the correct answer was among the options, but chose "none of the above" correctly only 7% of the time when it was not (Zhong et al., 2026).

A NOUL question about each option, in contrast, was 99% accurate on the same problems. A threshold chosen on separate data raised the rate of correct rejections to 79%, while keeping 97% accuracy when the answer was present. The model thus has the needed knowledge, but the CHOICE question does not express it.

Compressed scales. On ordered scales, Jev and open models of the same kind use only 67 to 76% of the range of levels that occur in the data, across 36 datasets (Gao et al., 2026). On unordered categories they use 87 to 102%. The compression grows as the scale becomes finer. On ANLI (Nie et al., 2020), Jev assigns 38.8% of its predictions and 51.3% of its errors to the middle category *neutral*. Targeted fine-tuning greatly reduces this compression, which shows that it is learned rather than built into the architecture. This agrees with the finding that Jev and LLM judges deviate from human ratings on ordered scales in the same way, mostly toward lower levels (Rao & Callison-Burch, 2026).

9.6 Limits of single-pass inference

Across studies, a recurring but not universal pattern appears. Typed decision models, most of which are single-pass, tend to perform worst relative to generative models when the answer must be worked out in several steps. Reported weaknesses include multi-step arithmetic and mathematics (Li et al., 2026g; Cheng et al., 2026b; Li et al., 2026i), judgements of code and logic (Li et al., 2026i), predicting consequences that are not stated (Yang et al., 2026b), specialist knowledge (Duan et al., 2026; Madrid-García & Merino-Barbancho, 2026), graph structure (Yang et al., 2026a), operating conditions in fault diagnosis (Jung, 2026), and numerical sensor features (Konak, 2026). Language and culture are a further factor. Performance drops on Chinese conversations (Li et al., 2026h) and on Chinese tasks in general (Wang et al., 2026c). Simulated cultural values match human differences much better in English than in Arabic (Asseri & Asseri, 2026). A Turkish benchmark adds tests based on translation and substituted names (Teke, 2026a). A preregistered study confirms the pattern for Spanish (marcosmartinez, 2026). An input in Spanish rather than English reduced accuracy by 3.0 to 6.4 points and roughly doubled the calibration error on harder tasks, while the language of the instructions made no difference.

Many decisions in sequence. Reliable single decisions do not guarantee a reliable system. On τ-bench (Yao et al., 2024), faster single decisions reduced the time per episode but also lowered task success, because errors accumulated over long sequences (Duan et al., 2026). In security, high overall accuracy can hide errors concentrated in certain attack types (Liu, 2026b). A further benchmark confirms this with puzzles checked by an exact solver (Exemocaro, 2026). Jev chose an optimal move in about 90% of positions, but solved only 57 of 100 puzzles, fewer than a reasoning LLM (80). Systems that make many typed decisions therefore need an error budget for the whole task, not only for each decision.

10 Discussion

10.1 A new format or a new ability?

In our reading, the main contribution of Jev is a new way of asking models for decisions, rather than an ability that general LLMs lack. General LLMs used through their next-token probabilities match openly released models of the same size (Li & Wagle, 2026). Reasoning can be distilled into a single first token, and the resulting models match Jev on a broad benchmark (Duan et al., 2026). An early review also found no independent accuracy advantage of typed answers (Tang & Zheng, 2026). What Jev has shown is that combining several known ideas changes how decision components are built: questions with declared answer types, independent answers to many questions, explicit probabilities and confidence scores, and fast, cheap inference. More than two thousand public software projects used Jev soon after its release (Ling et al., 2026), which suggests that this combination met a real need.

The evidence against this view deserves equal weight. Jev still leads the leaderboards on which it has been compared, although the best open reproduction is close (JevBench maintainers, 2026; apolinario, 2026). In a paired study it is significantly more accurate than the leading open encoder model on most types of agent decisions (Li, 2026b), and it outperforms a 27B open LLM on most of 37 datasets (Deußer et al., 2026).

Table 7: Failure modes of typed decision models and possible remedies.

Failure	Evidence	Possible remedy
Following names over definitions	(Sun et al., 2026; Azizi et al., 2026a)	test with neutral or contradictory names; present definitions only
Sensitivity to added context	(Xu et al., 2026b; Liu, 2026a; Hu et al., 2026)	treat the input as untrusted; check robustness to rewording
Sensitivity to option order	(Provatas & Georgakopoulos-Soares, 2026; Zhang et al., 2026b)	average over option orders, at least for uncertain inputs
Prompt injection	(Wu & Lim, 2026a; Wang et al., 2026a; Azizi et al., 2026b)	limit the allowed answers; use fixed rules for security decisions
Confidence does not show missing knowledge	(Shankaranarayana et al., 2026)	ask directly whether the evidence is sufficient
Inconsistent probabilities	(Li et al., 2026e; Joy, 2026; Porcedda, 2026)	check related questions for consistency
Probabilities depend on question type	(Zhong et al., 2026; Porcedda, 2026; Li et al., 2026e)	calibrate each question type separately
Several inputs in one request	(yodablocks, 2026)	one input per request when ranking
Missing answers not rejected	(Zhong et al., 2026)	use NOUL questions; tune thresholds
Compressed ordered scales	(Gao et al., 2026; Rao & Callison-Burch, 2026)	check use of all levels; targeted fine-tuning
Multi-step reasoning	(Li et al., 2026i; Yang et al., 2026b; Duan et al., 2026)	pass such inputs to a reasoning model
Same errors as the fallback	(Rao & Callison-Burch, 2026)	choose fallbacks that make different errors
Calibration changes with data	(Lu & Wu, 2026; Shankaranarayana et al., 2026; Nandakishore, 2026b)	recalibrate for each option set and domain

Matching Jev therefore requires either a strong base model or a careful correction of the readout (Li & Wagle, 2026; Zhang et al., 2026b). Moreover, no study has yet compared the calibration and robustness of Jev and the best open models under identical conditions. Our view is therefore limited: where it has been measured, the accuracy gap between Jev and comparable open LLMs is small. Further evidence on calibration or robustness may change this view.

10.2 A common question format

The typed question format now has several features of a standard. It has several independent implementations, including open reproductions and the models of Cloudflare and OpenAI. Models from different providers can answer the same questions (Cloudflare, 2026; bradAGI, 2026). Independent benchmarks compare them under the same conditions (apolinario, 2026; hotchpotch, 2026). On public sets of questions, the accuracy of open models is already statistically indistinguishable from that of Jev (bradAGI, 2026). If the format is the main innovation and the ability is widely available, then models will differ mainly in calibration for each question type, robustness, latency, and cost. These are also the properties for which the current evidence is weakest. A common format also raises a new question. The providers do not define confidence in the same way, and one provider explicitly recommends choosing thresholds from labeled examples rather than relying on the confidence score (OpenAI, 2026). The same threshold may therefore mean different things for different models.

10.3 Where the benefits are clear

Table 8 lists direct comparisons from academic studies between Jev and other approaches. We list them individually and do not aggregate them, because the studies differ in size, measures, baselines, and tasks, and a single study may cover one task or dozens. Read individually, the comparisons show a pattern that depends strongly on the comparator. Against generative LLMs used without training, typed decision models are often competitive, particularly where speed or cost matters most. Against supervised models trained on labeled data from the same domain, they are usually worse, as they are on tasks that require reasoning or subtle judgement. The clearest benefits are lower latency and cost in the reported settings, and more predictable behaviour. In network control, the share of requests handled within time limits changes fundamentally (Li et al., 2026c;d). In agents, typed decisions replace many calls to strong models (Wu & Lim, 2026b; Jiang et al., 2026; Zhang, 2026). In measurement, labeling hundreds of thousands of documents with probabilities becomes practical (Rafe & Das, 2026a). These cost advantages, however, depend on the deployment setting and can disappear, for example when generative models are used with discounted batch processing (Denney & DiGiuseppe, 2026). Gains in accuracy depend on the task, while losses on tasks that require reasoning are consistent across the studies we reviewed.

10.4 Combinations need models that make different errors

Combining a fast typed decision model with a slower fallback is the most common design. Its success requires more than passing uncertain inputs to the fallback. Proposition 1 shows that the fallback must correct more of the model's errors than it introduces, by a margin larger than its cost. Different errors are therefore necessary, but not sufficient. The gain is large when the two models fail on different inputs. An example is mathematics and code, where Jev fails but a reasoning judge succeeds (Li et al., 2026i). The gain is small when both models fail on the same inputs, as on ordered rubric criteria, where Jev and LLM judges make the same errors (Rao & Callison-Burch, 2026). Ideally, the routing score should therefore predict the fallback advantage $d(s)$ of Section 3.3, that is, where the fallback does better, rather than only whether the typed decision model is correct. This is a different goal from calibrating the model's own confidence, and it connects typed decision models to learning to defer. Rao & Callison-Burch (2026) likewise argue that future models should be designed with this goal in mind.

10.5 Typed decisions and reward models

Typed decision models and reward models are closely related. A reward model is analogous to a single fixed scoring or pairwise question, although it usually returns an unbounded scalar rather than a distribution over a finite answer space, and a policy is optimized against it. A typed decision model answers questions defined at inference time, and its answers are usually only read. Typed decision models are now also used as rewards, for image generation (Li et al., 2026a), to guide reinforcement learning (Ma et al., 2026c), and to train policies with verifiable rewards (Wen et al., 2026). They can therefore be expected to suffer from the same over-optimization as reward models (Gao et al., 2023). The weaknesses described in Section 9, such as answers changed by natural context or by added opinions, are exactly the weaknesses that an optimizer would exploit.

10.6 Cheap judgement at scale

When judgement becomes cheap, it is no longer used sparingly. Hill (2026) discusses the consequences for organizations and for knowledge when evaluation moves "from scarcity to abundance", and asks when cheap machine judgement can replace human assessment. The name Jev refers to the economist Jevons (Almeida, 2026; Jevons, 1865), who observed that greater efficiency can increase total consumption. Whether this happens for decisions is an empirical question that current evidence cannot answer. A technical consequence, however, is clear. An error rate that is acceptable for a few decisions produces many errors when applied to millions. Systematic biases, such as a preference for the middle category, for option names, or for salient options, are then repeated at scale rather than averaged out by human variety.

Table 8: Direct comparisons between Jev and other approaches in academic studies. Outcome for Jev on the main quality measure of each study: + better, ≈ comparable or mixed, − worse. Figures are as reported by the authors. Studies differ in size and design and should not be counted as equal votes.

Task	Compared with	Quality	Efficiency	
Agent-trace security (Wang & Gao, 2026)	4 generative judges	F1 77.8 vs. 74.1 (dataset-dependent)	∼$0.0002 per judgement	+
Alignment-failure detection (Guo et al., 2026)	supervised detectors; LLM scorers	median AUROC 0.886; beats supervised on most of 44 benchmarks	63× cheaper than LLM scorers	+
37 classification / reasoning datasets (Deußer et al., 2026)	27B open LLM	ahead on 27 of 37	–	+
Edge service orchestration (Li et al., 2026c)	fastest LLM	live success 0.91–0.95 vs. <0.1	22.7–64.5% lower latency	+
Open RAN intent (Li et al., 2026d)	hosted LLMs	1 s deadlines met on 99.8% vs. ≤17.9% of calls	–	+
Knowledge tracing, no target data (Lee & Park, 2026)	trained deep models	higher mean AUC (0.706)	far lower cost	+
Pairwise judging (Li et al., 2026i)	16 judges incl. GPT-6	within 3 points where readable; behind on math, code, logic	0.36% of fee	≈
Rubric judging (Rao & Callison-Burch, 2026)	3 flash-tier LLM judges	significant difference in ≤8 of 27 comparisons	16–325× cheaper	≈
Hate-speech moderation (Paschalides et al., 2026)	commercial LLM	within 1.6 macro-F1	97% lower cost	≈
Political-science replications (Denney & DiGiuseppe, 2026)	LLMs	comparable accuracy; better calibration	no gain at batch pricing	≈
Knowledge / reasoning / multilingual (Li et al., 2026g)	frontier LLMs	competitive on knowledge; −17.7 on MathQA	–	≈
CSS annotation, 18 tasks (Ibrahim & Zaki, 2026)	19 LLMs	trails best LLM on 14 of 15; median −11.6 F1	44× cheaper	−
Emotion classification (Zhang & Yu, 2026)	strongest LLM	62.9 vs. 67.3 macro-F1	lower latency and cost	−
Traffic classification (Xu & Mei, 2026)	trained tree ensembles	28–34% vs. 67–70% accuracy	–	−
Activity recognition (Konak, 2026)	supervised models	0.04–0.12 vs. 0.69–0.91 macro-F1	–	−

10.7 Relation to earlier surveys

Earlier surveys of LLM-as-a-Judge (Gu et al., 2024; Li et al., 2024) cover generative judges and their biases. They do not cover typed decision models, calibration as a property promised to users, the limits of single-pass inference, or systems that combine models by confidence. An early review of the first 28 studies on typed decision models proposed criteria for evaluating them (Tang & Zheng, 2026). Two working drafts maintained online also review this area. One focuses on calibration, selective control, and open implementations (Luo, 2026), and the other places Jev in a broader history of decision models since classical classifiers (Anonymous Authors, 2026). Our survey covers a much larger body of work, including models from several providers and open reproductions. It also adds a formal description, a classification of models and training methods, an analysis of efficiency, and a systematic account of failure modes.

11 Open Problems

When should a fast model pass a decision on? The evidence–computation boundary (Section 3) has been observed but cannot yet be predicted. Can a model tell from the input alone whether the answer can be read directly or must be worked out, and pass the input on in the second case? Work on switching between fast and slow models (Bailo, 2026), on repeated layers (Gan et al., 2026), and on the value of information (Yilmaz et al., 2026; Kang et al., 2026) offers starting points. A general theory would connect this question to adaptive computation and to the allocation of computation at inference time (Graves, 2016; Snell et al., 2024).

Calibration that holds in practice. Calibration matters most near the decision threshold, for the option set actually used, and under changes in the data. Conformal prediction and selective prediction give guarantees when training and test data come from the same distribution (Angelopoulos & Bates, 2021; Geifman & El-Yaniv, 2017; Jung et al., 2024). Typed decision models, however, face changing option sets (Lu & Wu, 2026), questions beyond their knowledge (Shankaranarayana et al., 2026), and adversarial inputs (Hu et al., 2026). Checks that need no labels, based on the stability of answers (Li & Chen, 2026b) or on the consistency of related answers (Li et al., 2026e), are promising because labels are rarely available after deployment.

Consistent answers to related questions. Evaluating each question separately is efficient, but nothing then enforces consistency between answers to related questions (Li et al., 2026e; Joy, 2026). Can models answer a set of related questions jointly, respecting constraints such as mutual exclusion or implication, while keeping the efficiency of parallel answers? Representing an explicit "unknown" outcome is one possible approach (Porcedda, 2026).

Following definitions rather than names. Users state their rules in option definitions, but models follow option names (Sun et al., 2026; Azizi et al., 2026a). Training objectives and ways of presenting options that make decisions independent of names are needed before typed decision models can be trusted to apply rules. Benchmarks that state the rule only in the definitions, such as PolicyBench, are a first step.

Robustness to untrusted input. The input to a typed decision model is often written partly by untrusted parties. Defences must limit how much an added opinion or natural context can change a decision (Xu et al., 2026b; Hu et al., 2026), and must treat decisions with a small margin as vulnerable (Wu & Lim, 2026a). Certified robustness for typed decision models, and protection of open models against poisoned training data (Wang et al., 2026a), are largely unexplored.

Combinations with different errors. Training a fast model and its fallback *together*, so that the fast model's confidence predicts where the fallback does better (Rao & Callison-Burch, 2026), connects to learning to defer (Madras et al., 2018; Mozannar & Sontag, 2020). It remains an open training problem.

Open and reproducible training. The training method of the most widely used model is not public. OpenJev-RLCD shows that reinforcement learning with proper scoring rules is feasible, but it relies on

generated rationales (Gao & Wang, 2026). An open single-pass model trained on observed outcomes, with documented data, would allow a controlled comparison with training on soft labels followed by calibration.

Images, tables, and graphs. Multimodal extensions (Zhou et al., 2026a; Yu & Yao, 2026; Ma et al., 2026a; Tang & Chen, 2026) show that the question format carries over to images, but calibration does not automatically follow. Graphs (Yang et al., 2026a), tables (Rafe & Das, 2026b), and sensor data (Konak, 2026) remain difficult.

Typed decisions as rewards. When typed decision models are used as rewards, the policy being trained may exploit their errors (Gao et al., 2023). It is unknown how quickly policies find such errors, and whether calibrated confidence can make optimization safer.

Comparability across models. Several providers now offer the same question types (TypeSafe AI, 2026b; Cloudflare, 2026; OpenAI, 2026). Comparing them requires shared definitions of confidence, ordered scores, "none of the above" options, and the grouping of questions, as well as calibration reported for each question type. Shared benchmarks (apolinario, 2026; hotchpotch, 2026) are a start, but none yet tests whether a threshold chosen for one model is valid for another.

General versus specialized models. Very small specialized models can outperform general typed decision models on their own tasks (Cua AI, 2026; Chen et al., 2026), and supervised models still perform better in several domains (Xu & Mei, 2026; Konak, 2026; Rafe & Das, 2026c). When is a general model worth its breadth? And can it be used to train specialized models, through distillation (Sadoun et al., 2026) or reinforcement learning, with guarantees on their quality?

Effects on society and institutions. Cheap and confident decisions without explanations, made at large scale, raise questions of accountability, of the right to contest decisions, and of fairness across languages and cultures (Asseri & Asseri, 2026; Li et al., 2026h). As evaluation becomes abundant (Hill, 2026), norms are needed for when a typed decision may stand without human review.

12 Conclusion

Many applications of language models need decisions rather than text. Typed decision models address this need directly. They answer questions with predefined answer types, return probabilities over the possible answers, and, for some question types, a confidence score, all without generating text. They build on a long line of earlier work, and Jev made the approach widely known. A large and varied body of research has since studied these models in judging, routing, agent control, security, database queries, and large-scale annotation. Several providers and many open projects now use the same question format.

The findings are nuanced. In the reported settings, typed decision models are often much faster and cheaper than generative models, although this advantage depends on the deployment setting. Their outputs are always in the declared format. They are often competitive with generative LLMs used without training when the needed evidence is stated in the input. They tend to perform worse on tasks that require multi-step reasoning, and they usually trail supervised models trained on in-domain data. Their probabilities and confidence scores are useful for deciding when to pass an input to a stronger model, but they are not calibrated in all settings and do not reveal missing knowledge. A valid format does not protect against following option names, against misleading context or injected text, against inconsistent probabilities, or against failures on missing answers and ordered scales. Combining a typed decision model with a fallback improves accuracy only if, on the inputs passed to it, the fallback corrects more errors than it introduces. Much of the underlying ability already exists in general LLMs, so the lasting contribution may be the question format itself. Making typed decisions reliable requires models that follow definitions, routing scores that identify where a fallback helps, consistent answers to related questions, robustness to untrusted input, and open and reproducible training. These are the main directions for future research.

Broader Impact Statement

Typed decision models make automated judgement much cheaper. This can widen access to useful applications, such as content moderation, the screening of requests, assistive tools, and large-scale measurement for research and public policy. The same property also creates risks, which this survey describes. Decisions are made without an explanation that people can inspect. Confident errors can be caused by natural context, by untrusted text, or by misleading option names. Probabilities are not equally well calibrated across question types, languages, and cultures. Cheap decisions may be used at a scale where systematic biases are repeated rather than averaged out. Typed decision models can also be used both to detect and to carry out misuse. We therefore recommend that applications treat the model's input as untrusted and keep fixed rules and human review for decisions that affect security or people's rights. Calibration should be reported for each question type and language, and people should be able to contest automated decisions. This survey reviews publicly available work. It does not introduce new models, data, or experiments with human participants.

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A Literature Collection

Scope. We review studies of typed decision models together with the earlier work needed to place them in context (Section 2). This survey is a structured review rather than a formal systematic review. We document how studies were found and selected so that the selection can be checked and repeated.

Sources and queries. We searched arXiv through its public interface, because nearly all studies in this area were available there as preprints. We used one primary query and six supplementary queries over all fields:

- primary: `Jev`;
- supplementary: `"System One"`, `"System-One"`, `"typed decision"`, `"decision model" AND typed`, `RLCD`, and `"System-1" AND decision`.

The supplementary queries were restricted to submissions between 15 September 2026, the public release of Jev, and 8 October 2026, the last submission date covered. All queries were run in October 2026 and re-run with the same date limits to confirm the counts below.

Selection. Figure 2 summarizes the selection. The primary query returned 102 records. Two were excluded on title and abstract because they are unrelated: a finance paper by an author named Jev, and an earlier paper on document reranking. The six supplementary queries returned 161 records, of which 85 were unique and 30 were not already found by the primary query. Of these 30, six were excluded on title and abstract because they use the search terms in an unrelated sense (four papers in physics, mathematics, and astronomy, one on control engineering, and one on the stability of fluid flows). One further record was read in full and excluded under criterion (ii) below, because the described system does not contain a learned model: its decisions are generated from coordinates on the boundary of the Mandelbrot set. The remaining 23 records were included. Following references in both directions from the included studies added six further studies. In total, the survey covers 129 studies. Appendix B lists them by topic.

Inclusion criteria. A study is included if (i) it proposes, trains, evaluates, attacks, or applies a model that answers questions with predefined answer types and returns probabilities over the answers, or if it studies Jev or a model explicitly presented as similar to it, and (ii) the model is learned from data, either trained or adapted from a pretrained model. Studies that only mention Jev are included only if they report results for such a model.

Other sources. Some evidence on typed decision models is published outside academic venues. We cite documents from model providers only to report what the providers state. We cite other public sources, such as code repositories and online benchmarks, only if they report quantitative results together with released data, code, or model outputs. We found them through two public lists (awesome-jev contributors, 2026; heyjunpenn and contributors, 2026) and a public leaderboard (apolinario, 2026), and list them in Appendix C. These sources complement the academic studies; no general conclusion in this survey rests on them alone.

Synthesis. For each study we recorded the models evaluated, the baselines, the question types, the tasks, the main measures, and any reported costs or speeds. We report numbers as stated by the authors and did not repeat any experiments. Where studies disagree, for example on the direction of miscalibration, we report the disagreement and, where possible, a likely explanation. Table 8 lists direct comparisons individually. We do not aggregate them, because the studies differ in size, measures, and baselines.

Limitations. The review has several limitations. First, the corpus is drawn from a single source, arXiv, complemented by reference tracing; studies published only elsewhere may be missing. Second, the primary query uses the name of a specific model. This may over-represent studies that use Jev's terminology and miss studies of the same kind of model under other names. The generic supplementary queries, the reference tracing, and the review of earlier work in Section 2 reduce but do not remove this bias. Third, most included studies are preprints that had not completed peer review, commercial models may change between versions, and early studies may over-report positive results. Fourth, the studies were selected and summarized by the authors of this survey without independent double coding, and we did not perform a formal assessment of the risk of bias of each study. Instead, we report figures exactly as published, point out single runs, small samples, and model-generated reference labels where they occur, and present evidence against each main claim alongside the evidence for it.

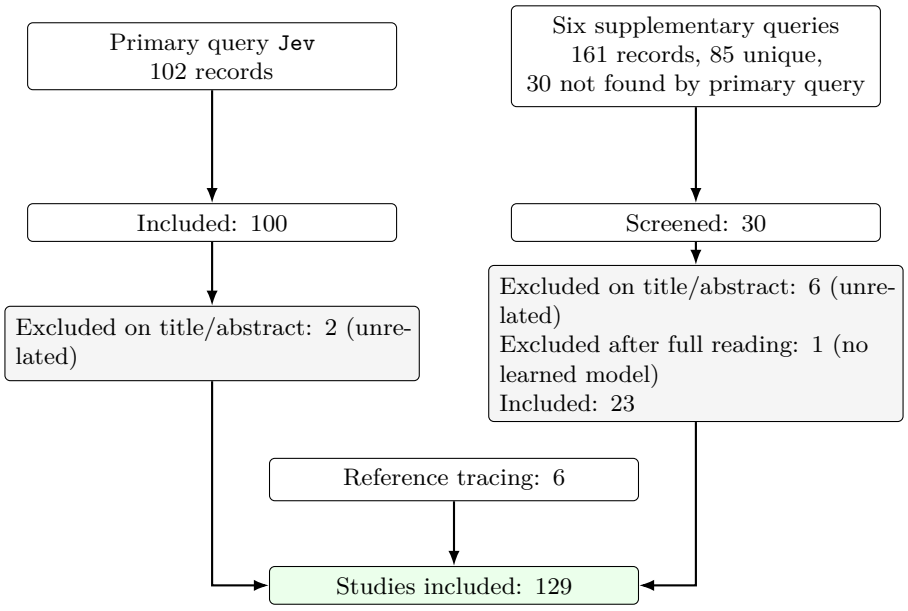


Figure 2: Selection of studies. Counts refer to arXiv records submitted up to the last covered date.

B List of Surveyed Studies

Table 9 lists the 129 studies covered by this survey, grouped by the subsection that discusses each study most often. Full references are given in the bibliography.

Table 9: Studies covered by this survey, by topic.

Topic (section)	Studies
Jev (§4)	Li & Chen (2026b)
Open models designed for typed decisions (§4)	dos Santos Barbosa (2026); Nandakishore (2026b); Ding et al. (2026); Amit (2026); Deka (2026); Chen et al. (2026); Teke (2026b); Gan et al. (2026); Huo et al. (2026)
General LLMs as typed decision models (§4)	Ye et al. (2026); Tang & Zheng (2026); Ding (2026); Dai et al. (2026); Zhan et al. (2026); Li & Wagle (2026); Wu et al. (2026a); Sonthalia et al. (2026); Sadoun et al. (2026)
Open reproductions (§4)	Nandakishore (2026a)
Multimodal models (§4)	Zhou et al. (2026a); Qin et al. (2026); Sheng et al. (2026); Tang & Chen (2026); Cheng et al. (2026a)
Training objectives (§5)	Wen et al. (2026); Lin (2026); Gao & Wang (2026); Ma et al. (2026c)
Calibration after training (§5)	Li (2026a); Lu & Wu (2026)
Sources of speed (§6)	Cheng et al. (2026b); Yu & Yao (2026); Wu & Lim (2026b); Provatas & Georgakopoulos-Soares (2026); Shi et al. (2026); Wang et al. (2026c); Rajput et al. (2026); Zhang et al. (2026b); Wang et al. (2026d); Cao et al. (2026); Li & Chen (2026a); Wake & Wagle (2026); Zhou et al. (2026b); Ma et al. (2026a)
Judging and evaluation (§7)	Robitza (2026); Huang et al. (2026); Rao & Callison-Burch (2026); Liu (2026b); Wang & Gao (2026); Loi (2026); Xu et al. (2026a); Li et al. (2026a)
Routing and network control (§7)	Li et al. (2026c;d); Kwartler et al. (2026); Wang et al. (2026b); Kourtis & Xilouris (2026); Xu & Mei (2026); Rahimi et al. (2026); Li et al. (2026b); Kim (2026); Besanson (2026)
Agent control (§7)	Jiang et al. (2026); Ma et al. (2026b); Zhang (2026); Sharma & Lall (2026); Liu et al. (2026); Wang (2026); Su & Huang (2026); Wu et al. (2026b); Bailo (2026); Fan et al. (2026)

Topic (section)	Studies
Safety, security, and moderation (§7)	Rezvani et al. (2026); Severo et al. (2026); Mu et al. (2026); Paschalides et al. (2026); Mazzoni & Failla (2026)
Large-scale annotation and measurement (§7)	Rafe & Das (2026a); Ibrahim & Zaki (2026); Deng et al. (2026); Zhang et al. (2026a); Yilmaz et al. (2026); Cinà (2026); Madrid-García & Merino-Barbancho (2026); Zhang et al. (2026c); Konak (2026); Asseri & Asseri (2026); Lyu & Xia (2026); Rafe & Das (2026d;c); Denney & DiGiuseppe (2026); Kang et al. (2026); Zhang & Yu (2026); Li et al. (2026f); Jung (2026); Rafe & Das (2026b); Yan (2026); Lee & Park (2026); Li et al. (2026h)
What should be measured (§8)	Joy (2026); Chen & Li (2026); Sharma et al. (2026)
Benchmarks (§8)	Ren et al. (2026); Guo et al. (2026); Porcedda (2026); Deußer et al. (2026); Teke (2026a); Azizi et al. (2026a); Yang et al. (2026a); Li et al. (2026g)
Common problems in current evaluations (§8)	Hu et al. (2026); Li (2026b)
Following option names instead of definitions (§9)	Sun et al. (2026)
Sensitivity to context and option order (§9)	Xu et al. (2026b); Liu (2026a)
Prompt injection and security (§9)	Wu & Lim (2026a); Wang et al. (2026a); Azizi et al. (2026b)
Calibration and consistency of probabilities (§9)	Li et al. (2026e); Shankaranarayana et al. (2026)
Missing answers and ordered scales (§9)	Gao et al. (2026)
Limits of single-pass inference (§9)	Li et al. (2026i); Zhong et al. (2026); Yang et al. (2026b); Duan et al. (2026)
When is a single pass enough? (§3)	Xue et al. (2026)
Roles of decisions (§3)	Ling et al. (2026)
Cheap judgement at scale (§10)	Hill (2026)

C Other Public Sources

Table 10 lists the public sources outside academic venues that are cited in this survey. For each source, it gives what was measured, the sample size, and the measures taken to make the results reliable. Lists and model repositories are cited only to describe available models or their stated properties. Quantitative statements in the text rely on the evaluation studies in the lower part of the table. None of these sources has been peer reviewed.

Table 10: Public sources outside academic venues cited in this survey.

Source	Content	Scale	Reliability measures
Lists and leaderboards			
awesome-jev contributors (2026)	curated list of projects	several hundred entries	explicit inclusion rules
heyjunpenn and contributors (2026)	list of models, tools, and benchmarks	nearly 1,000 entries	notes for each entry
apolinario (2026)	leaderboard of typed decision models	37 public benchmarks	fixed benchmarks; no prompt tuning; code released
JevBench maintainers (2026)	leaderboard with a combined score	534 decisions, 52 systems (v1.3.0)	public evaluation code

Source	Content	Scale	Reliability measures
Open models and data (stated properties only)			
jaredpalmer (2026), TokenRhythm (2026), Bespoke Labs (2026), denis-pplx (2026), Contrastive-LM (2026), TianyuCodings (2026), wfzyx (2026), Manavarya09 (2026), Xiaooolong (2026)	open Jev-style models	–	weights or code released; figures stated by authors
Cua AI (2026)	small model for form filling	196 real decisions on three forms	synthetic training data stated
sileod (2026)	training data in typed format	over 500K rows	options shuffled; sources recorded
Evaluation studies			
bradAGI (2026)	open model versus Jev on public questions	256 items	paired statistical test
scienthoon (2026)	calibration by question type on new data	900 synthetic + 3,721 public items	raw outputs released
Hayashi (2026)	probabilities for known chances	400 trials + forecast documents	raw outputs and code released
jjd-lab (2026)	question type in simulated surveys	24,596 paired cells	criteria fixed in advance
Exemocaro (2026)	single moves versus whole tasks	100 puzzles; 495 positions	exact solver
yodablocks (2026)	ranking by probability	360 + 306 labelled rows	preregistered criteria
marcosmartinez (2026)	language of input and instructions	3,200 paired items	preregistered paired design
hotchpotch (2026)	broad benchmark	137 benchmarks	public data
anessbelbati (2026)	reranking	14 datasets	raw outputs; confidence intervals
nikkoxgonzales (2026)	statistical guarantees for thresholds	routing benchmark	checked on new data
nican2018 (2026)	monitoring of agent sabotage	13 attacks	thresholds checked on separate tasks
Gaurav-Gosain (2026)	predicting results of headline tests	10,984 pairs	both option orders tested
Shea & Roche (2026)	typed versus generative judges	5 cases × 100 repetitions	human labels; small sample